Sri Lanka Institute of Information Technology

IT3021 – Data Warehousing and Business Intelligence

Assignment 01

MADUGALLE K J W R E W N M R O D

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Kandy Uni

Task 1: Data Set Selection

1. Description of the Selected Dataset

The selected dataset is a **Hotel Sales and Booking OLTP system** containing **6,050 individual transactions** recorded throughout the year **2024**. It functions as a comprehensive operational source, capturing 37 distinct attributes (columns) for each booking. The data covers a wide range of business functions, from front-desk logistics to back-office financial reporting.

2. Attribute Breakdown

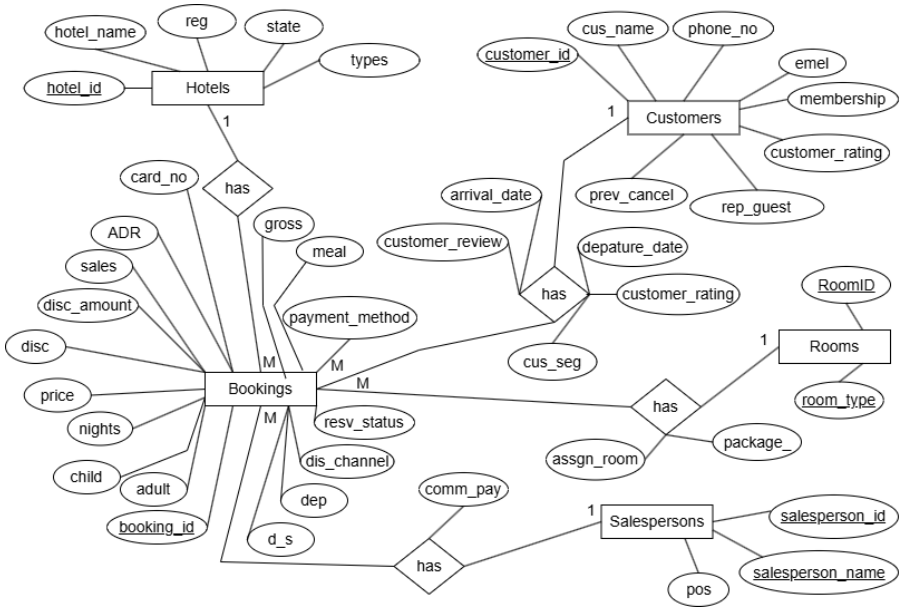
The 37 columns are categorised into logical business domains, providing a rich foundation for a multi-dimensional data warehouse:

- **Hotel Metadata:** Hotel Name, Type, Region, and State.
- **Customer Profile:** Name, contact details, "Repeated Guest" status, and historical cancellations.
- **Stay Logistics:** Check-in/out dates, guest counts (Adults/Children), and Room Type.
- **Financial Metrics:** Gross Sales, Net Sales, Discounts, Commission, and Deposit Status.
- **Performance & Feedback:** Customer Ratings, Reviews, and Sales Person details.

3. Justification for Selection

- **Data Volume:** With over **6,000 records** and a full year of data, it meets the requirement for sufficient volume to demonstrate ETL, SSAS cubes, and hierarchies.
- **Analytic Depth:** The 37 attributes allow for the creation of multiple dimensions (Guest, Hotel, Room, Salesperson) and a robust fact table with diverse measures.

4. ER-Diagram (ERD)



Task 2: Preparation of Data Sources

1. Data Source Strategy

To demonstrate the ability to integrate heterogeneous data, the hotel dataset has been prepared across three distinct source types. This mimics a real-world scenario where master data is stored in a relational database, while transactional and customer feedback data are managed via spreadsheets and flat files.

2. Source Format and Information Mapping

The following table explains exactly what information is available in each format:

Source Type	Physical Source	Information Available
Relational (SQL Server)	Hotels Table	Infrastructure details, including Hotel ID, Name, Region, State, and Type.
Relational (SQL Server)	Rooms Table	Inventory metadata: Room ID and Room Type.
Relational (SQL Server)	Salesperson Table	Staff metadata: Salesperson ID, Name, and Position.
Excel (.xlsx)	booking_data.xlsx	Core transactional logistics: Check-in/out dates, guest counts, and full financial metrics (Gross Sales, Discounts, Net Sales, etc.).
Flat File (.csv)	customer_data.csv	Guest profile and sentiment data: Name, Phone, Email, Ratings, Reviews, and Loyalty status.

3. Detailed Column Availability

- **SQL Server (Hotel_OLTP_Source):** * *Columns:* hotel_id, hotel_name, reg, state, types, RoomID, room_type, salesperson_id, salesperson_name, pos.
- **Excel (booking_data.xlsx):** * *Columns:* customer_id, hotel_id, room_id, salesperson_id, resv_status, arrival_date, depature_date, nights, adult, child, dis_channel, gross, disc, disc_amt, sales, dep, d_s, ADR, payment_method, card_no, comm_pay, assgn_room, cus_seg, package_, price, meal.

- **CSV (customer_data.csv):** * *Columns:* customer_id, cus_name, phone_no, emel, membership, customer_rating, customer_review, rep_guest, prev_cancel.

4. Required Screenshots for the Report

To make your documentation "reproducible" (as per Task 6 requirements), include these:

1. **Screenshot of SSMS:** Showing the Hotels, Rooms, and Salesperson tables under your Hotel_OLTP_Source database.

120 %

Results | Messages

	hotel_id	hotel_name	reg	state	types
1	1	Le Meridien Sabah	East	Sabah	Resort Hotel
2	2	The Hilton Kuala Lumpur	Central	Kuala Lumpur	City Hotel
3	3	Lexis Suites Penang	North	Penang	City Hotel

	RoomID	room_type
1	1	Deluxe Room
2	2	Single Room
3	3	Studio
4	4	Royal Suite

	salesperson_id	salesperson_name	pos
1	SP-001	Kelly	Sales Executive
2	SP-002	Amy	Sales Executive
3	SP-003	Derek	Sales Executive
4	SP-004	Kelly	Sales Manager
5	SP-005	Leo	Sales Manager
6	SP-006	Bernice	Sales Executive

2. **Screenshot of Excel:** A view of the booking_data.xlsx file showing the first few rows of headers and data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
1	booking_i	customer	hotel_id	room_id	salespers	resv_stat	arrival_date	departure_dat	nights	adult	child	dis_chann	gross	disc	disc_amt	sales	dep	d_s	ADR	payment_	card_no	comm_pa	assgn_roc	cus_seg	package_	price	meal
2	BK-001	CUS-001	3	2	SP-001	Check-Ou	2024-04-02	2024-04-10	8	2	-	Direct	4,400	10%	440	4,280	300	Refundec	550	Credit Car	**********4322	15	C	Individual	320	550.00	BB
3	BK-002	CUS-002	3	1	SP-002	Check-Ou	2024-07-13	2024-07-19	6	2	-	Direct	3,900	5%	195	3,945	300	Refundec	650	Credit Car	**********9157	20	C	Individual	240	650.00	BB
4	BK-003	CUS-003	3	2	SP-003	Check-Ou	2024-08-22	2024-09-04	13	1	-	Agoda	7,150	10%	715	6,695	300	Refundec	550	Online Tra	**********3734	15	C	Corporate	260	550.00	BB
5	BK-004	CUS-004	1	4	SP-001	Check-Ou	2024-05-16	2024-05-22	6	3	1	Direct	10,800	10%	1,080	10,200	300	Refundec	1,800	Credit Car	**********5677	30	A	Family	480	1800.00	BB
6	BK-005	CUS-005	2	1	SP-001	Check-Ou	2024-02-05	2024-02-10	5	2	2	Direct	3,250	10%	325	3,325	300	Forfeited	650	Online Tra	**********5498	20	A	Family	400	650.00	BB
7	BK-006	CUS-006	3	2	SP-002	Check-Ou	2024-09-13	2024-09-27	14	1	-	Direct	7,700	5%	385	7,595	300	Refundec	550	Credit Car	**********9263	15	A	Corporate	280	550.00	BB
8	BK-007	CUS-007	2	2	SP-004	Check-Ou	2024-09-12	2024-09-17	5	1	-	Agoda	2,750	5%	138	2,713	300	Refundec	550	Online Tra	**********6994	15	C	Corporate	100	550.00	BB
9	BK-008	CUS-008	1	2	SP-005	Check-Ou	2024-09-15	2024-09-30	15	1	2	Booking.c	8,250	5%	413	8,738	300	Refundec	550	Online Tra	**********8729	15	C	Family	900	550.00	FB
10	BK-009	CUS-009	2	1	SP-001	Check-Ou	2024-06-24	2024-07-05	11	2	-	Booking.c	7,150	5%	358	7,233	300	Refundec	650	Online Tra	**********3639	20	A	Individual	440	650.00	BB

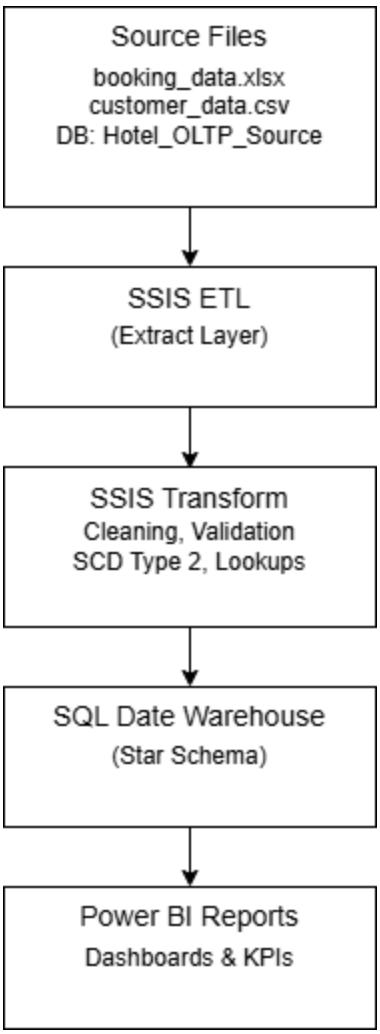
3. **Screenshot of Notepad:** Opening customer_data.csv in Notepad to show the raw comma-separated values.

```
customer_id,cus_name,phone_no,emel,membership,customer_rating,customer_review,rep_guest,prev_cancel
CUS-001,Ernest Barnes,669-792-1661,Ernest.Barnes31@outlook.com,Gold,5,Excellent,0,0
CUS-002,Andrea Baker,858-637-6955,Andrea_Baker94@aol.com,Gold,4,Very Good,0,0
CUS-003,Rebecca Parker,652-885-2745,Rebecca_Parker@comcast.net,Platinum,5,Excellent,0,0
CUS-004,Laura Murray,364-656-8427,Laura_M@gmail.com,Platinum,4,Very Good,0,0
CUS-005,Linda Hines,713-226-5883,LHines@verizon.com,Platinum,5,Excellent,0,0
CUS-006,Jasmine Fletcher,190-271-6743,JFletcher43@xfinity.com,Gold,5,Excellent,0,0
CUS-007,Dylan Rangel,420-332-5209,Rangel.Dylan@comcast.net,Gold,5,Excellent,0,0
CUS-008,William Velez,286-669-4333,Velez_William@mail.com,Gold,4,Very Good,0,0
CUS-009,Steven Murphy,341-726-5787,Steven.Murphy54@aol.com,Gold,5,Excellent,0,0
CUS-010,Michael Moore,316-648-6176,MichaelMoore81@outlook.com,Gold,4,Very Good,0,0
```

Task 3: Solution Architecture

3.1 Architectural Diagram

The diagram below illustrates the high-level architecture of the Hotel Data Warehouse and Business Intelligence solution. It follows a multi-tier approach to ensure data integrity, starting from heterogeneous sources through to the final reporting layer.



3.2 Component Descriptions

The following components represent the logical flow and processing of data within the system:

- **Heterogeneous Data Sources (OLTP Layer):**

The foundation of the solution consists of three distinct source formats to demonstrate multi-source integration capabilities:

  - **SQL Server Database:** Stores relational master data, including Hotel infrastructure, Room definitions, and Salesperson details.
  - **Excel Workbook (booking_data.xlsx):** Captures the core 6,050 transactional records for 2024, including arrival/departure dates and financial measures.

- **CSV Flat File (customer_data.csv):** Stores guest-specific profiles, loyalty status, and subjective customer feedback (ratings and reviews).

- **Staging Area:**

A dedicated landing zone within SQL Server where raw data is extracted before processing. This acts as a buffer to ensure that heavy ETL operations do not affect the performance of the live operational source systems.

- **ETL Layer (Microsoft SSIS):**

SQL Server Integration Services (SSIS) serves as the integration engine. It performs the "Extract, Transform, and Load" process by:

- Cleaning and validating raw data from the staging area.
- Executing **Data Conversions** to standardise data types (e.g., currency and date formats).
- Performing **Lookups** to map natural business keys to the surrogate keys required for the Star Schema.

- **Data Warehouse (Storage Layer):**

The processed data is loaded into a centralised SQL Server Data Warehouse designed as a **Star Schema**. This consists of a central **FactBookings** table (containing quantitative measures like Gross Sales, Net Sales, and ADR) surrounded by five descriptive dimensions: **DimHotel**, **DimGuest**, **DimRoom**, **DimSalesperson**, and **DimDate**.

- **Presentation Layer (Power BI):**

The final component of the BI solution. Power BI connects to the Data Warehouse to provide an interactive visualisation layer. It transforms the stored data into meaningful KPIs such as **Occupancy Trends**, **Revenue Performance**, and **Customer Satisfaction Analytics** for decision-makers.

Task 4: Data Warehouse Design & Development

4.1 Dimensional Model Description

The Data Warehouse is implemented using a **Star Schema** architecture. This design centralises quantitative business metrics into a Fact table, which is then contextually described by multiple Dimension tables. This structure is optimised for high-performance querying and intuitive reporting in Power BI.

Central Fact Table: FactBookings

The FactBookings table captures the details of 6,050 hotel transactions. It contains Foreign Keys (FK) pointing to the dimensions and various measures for analysis.

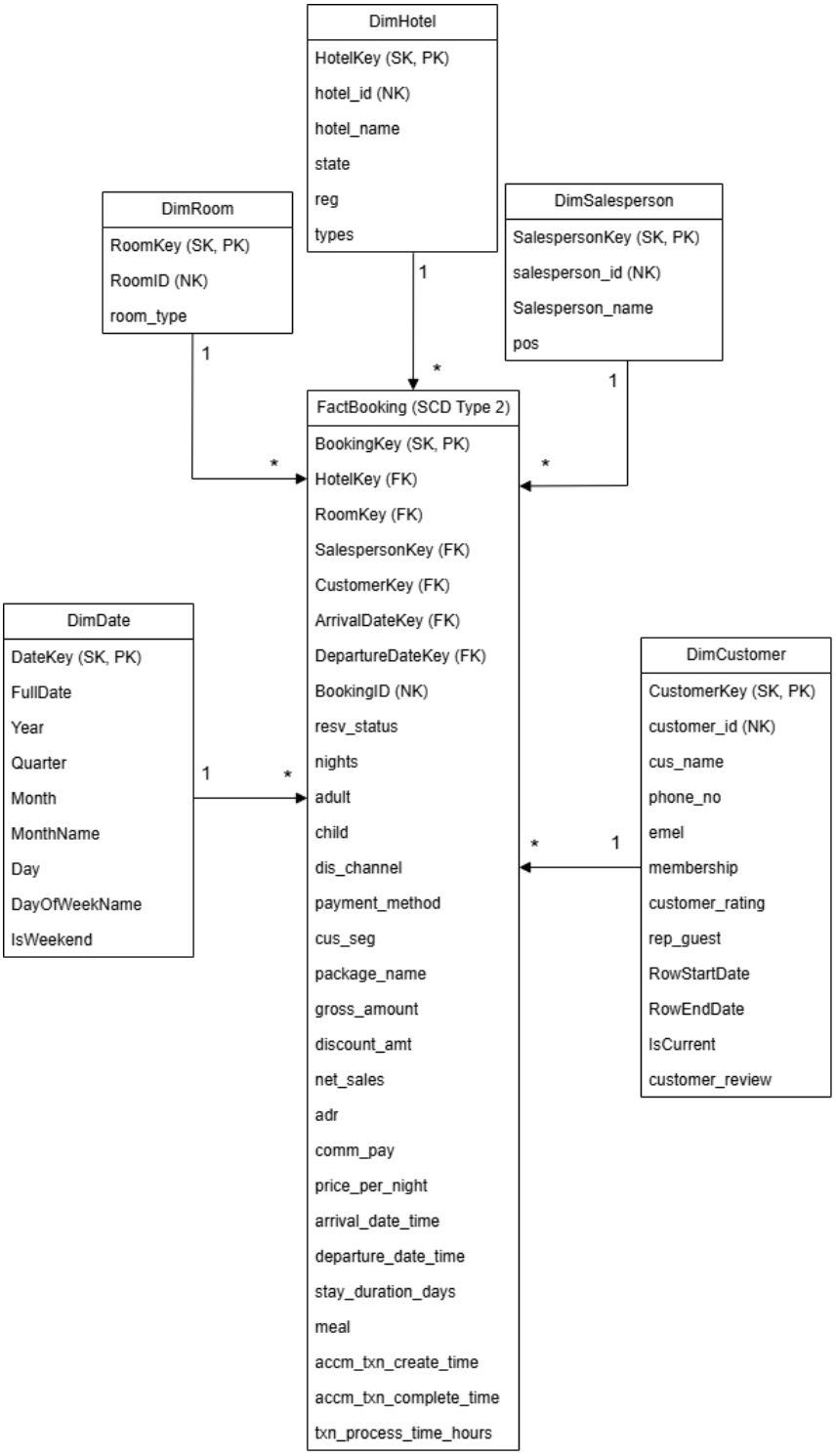
- **Key Measures:** gross_amount, discount_amt, net_sales, adr, comm_pay, price_per_night, and stay_duration_days.
- **Process Metrics:** txn_process_time_hours (calculated as the difference between accm_txn_create_time and accm_txn_complete_time).
- **Foreign Keys:** HotelKey, RoomKey, SalespersonKey, CustomerKey, ArrivalDateKey, and DepartureDateKey.

Dimension Tables:

- **DimCustomer:** Stores guest details. This table implements **Slowly Changing Dimension (SCD) Type 2** (indicated by RowStartDate, RowEndDate, and IsCurrent) to track changes in customer attributes like membership or phone_no over time.
- **DimDate:** A comprehensive time dimension allowing for analysis by Year, Quarter, MonthName, and DayOfWeekName. It includes an IsWeekend flag for specialised hospitality insights.
- **DimHotel:** Contains infrastructure data such as hotel_name, reg (Region), state, and types.
- **DimRoom:** A simple dimension capturing room_type linked to the specific RoomID.
- **DimSalesperson:** Tracks the performance of staff members, including their salesperson_name and pos (Position).

4.2 Relational Diagram (Star Schema)

The following diagram illustrates the relationships between the Fact and Dimension tables. The FactBookings table is at the centre, representing the many-to-one relationship with each dimension.



4.3 Design Assumptions

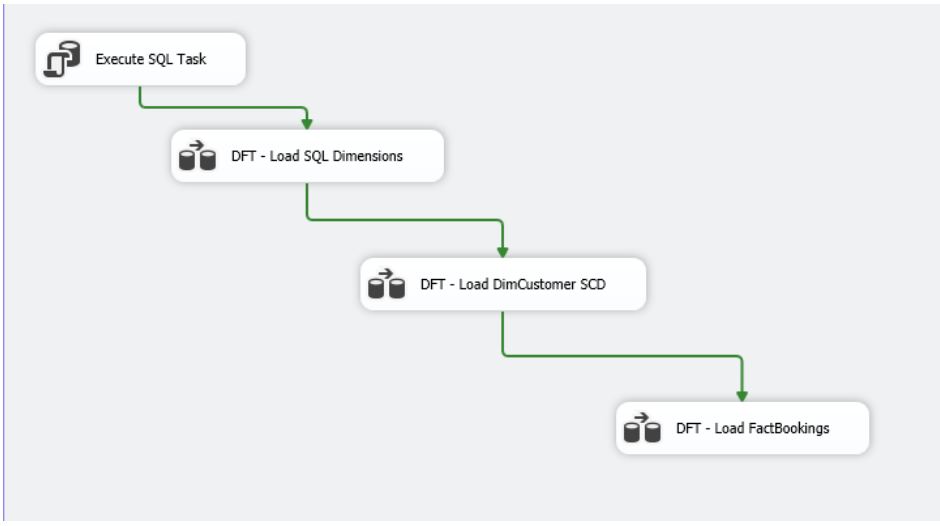
To ensure a robust and scalable BI solution, the following assumptions were made:

1. **Surrogate Keys (SK):** All dimension tables utilise system-generated integer Primary Keys (e.g., CustomerKey, HotelKey) instead of natural business keys (e.g., customer_id). This shields the Data Warehouse from changes in the source system's indexing.
2. **Tracking Historical Changes:** It is assumed that customer information (like membership level) is dynamic. Therefore, **SCD Type 2** was implemented in DimCustomer to maintain a historical record of guest profiles rather than overwriting them.
3. **The Grain of the Fact Table:** The grain is defined as **"One row per individual hotel booking ID."** This is the lowest level of transactional data available in the source.
4. **Date Handling:** Two separate Date Keys (ArrivalDateKey and DepartureDateKey) are used to link to a single DimDate table. This "Role-Playing Dimension" approach allows us to filter the same report by either the check-in or check-out date.
5. **Calculated Measures:** Fields such as net_sales and txn_process_time_hours are calculated during the ETL process to reduce the processing load on the reporting layer.

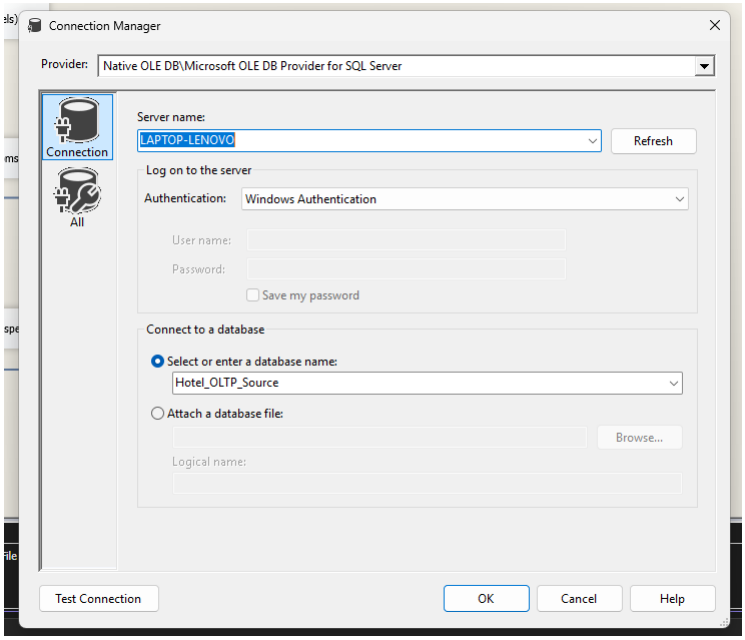
Task 5: Development of the Data Warehouse

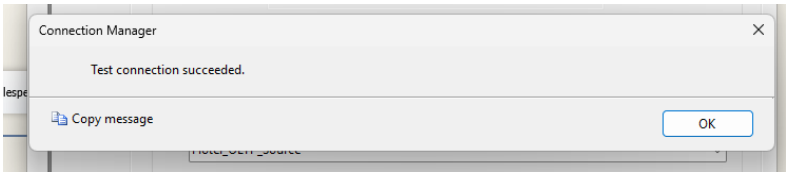
5.1 ETL Control Flow Implementation

The orchestration of the ETL process is managed through **SQL Server Integration Services (SSIS)**. The Control Flow is designed with a sequential dependency to ensure data integrity, particularly for the generation and retrieval of Surrogate Keys.

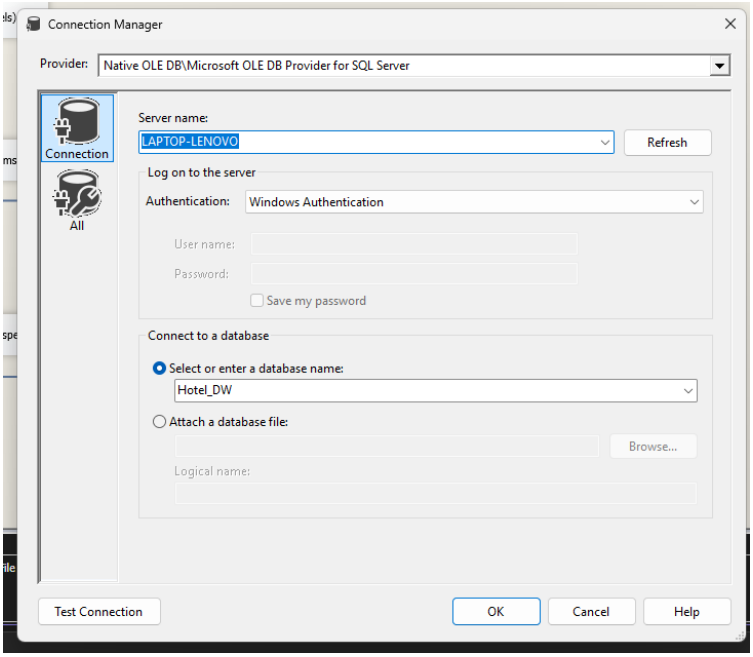


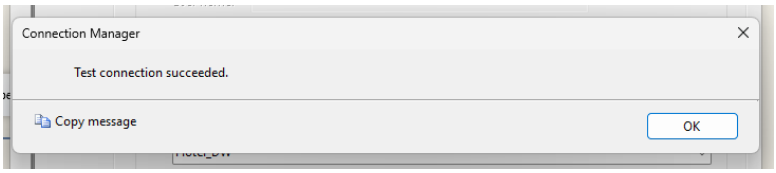
Connection Manager: Hotel_OLTP_Source





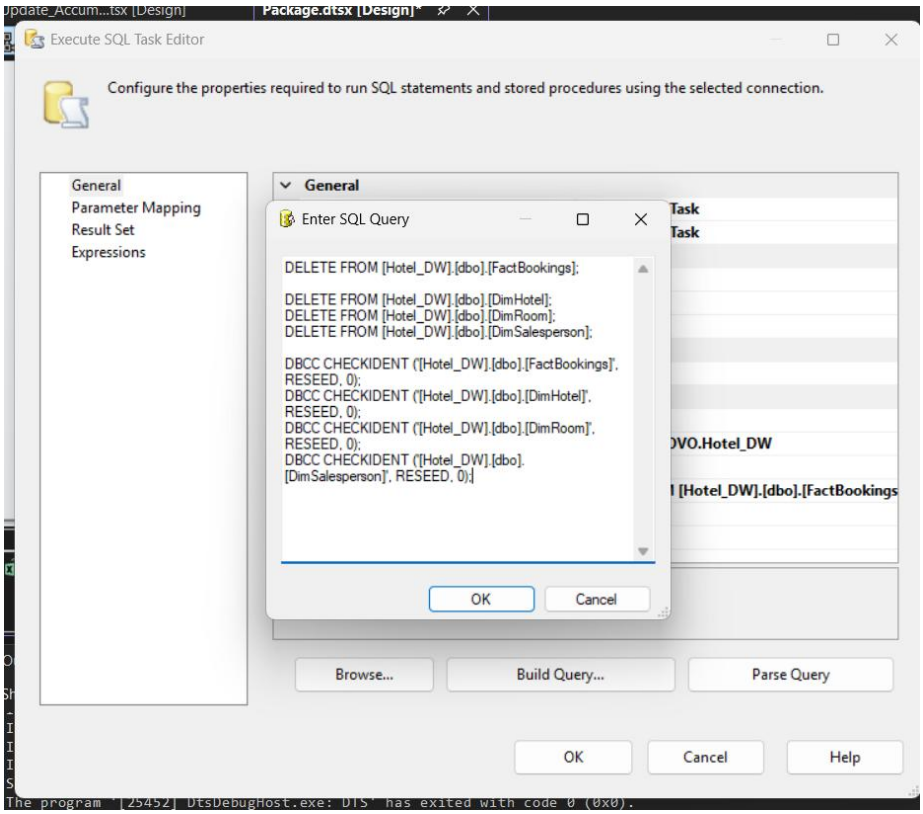
Connection Manager: Hotel_DW





Sequence Description:

1. **Execute SQL Task:** An Execute SQL Task that clears the FactBookings, DimHotel, DimRooms, DimSalesperson tables to prevent duplicate records during full loads.



DELETE FROM [Hotel_DW].[dbo].[FactBookings];

DELETE FROM [Hotel_DW].[dbo].[DimHotel];

DELETE FROM [Hotel_DW].[dbo].[DimRoom];

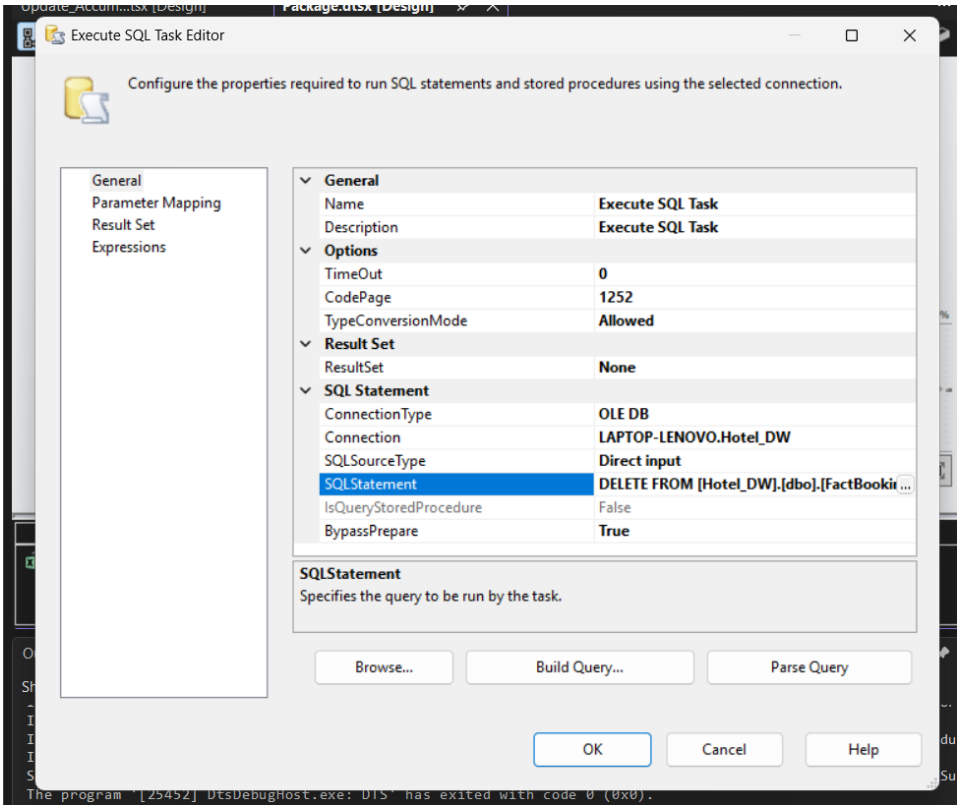
DELETE FROM [Hotel_DW].[dbo].[DimSalesperson];

DBCC CHECKIDENT ('[Hotel_DW].[dbo].[FactBookings]', RESEED, 0);

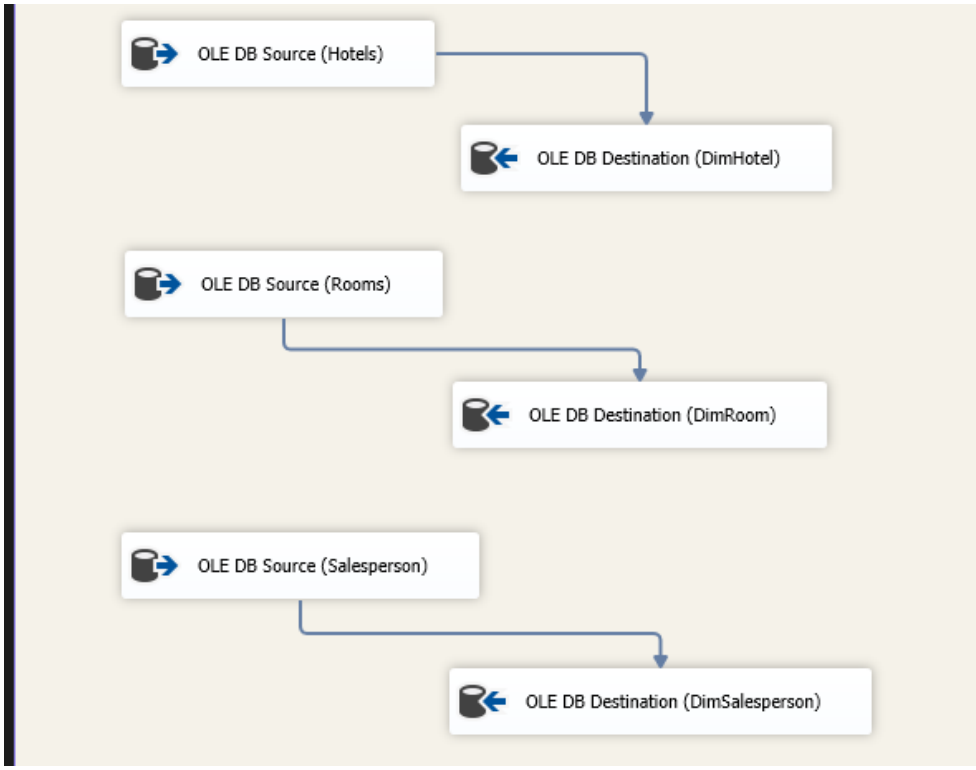
DBCC CHECKIDENT ('[Hotel_DW].[dbo].[DimHotel]', RESEED, 0);

DBCC CHECKIDENT ('[Hotel_DW].[dbo].[DimRoom]', RESEED, 0);

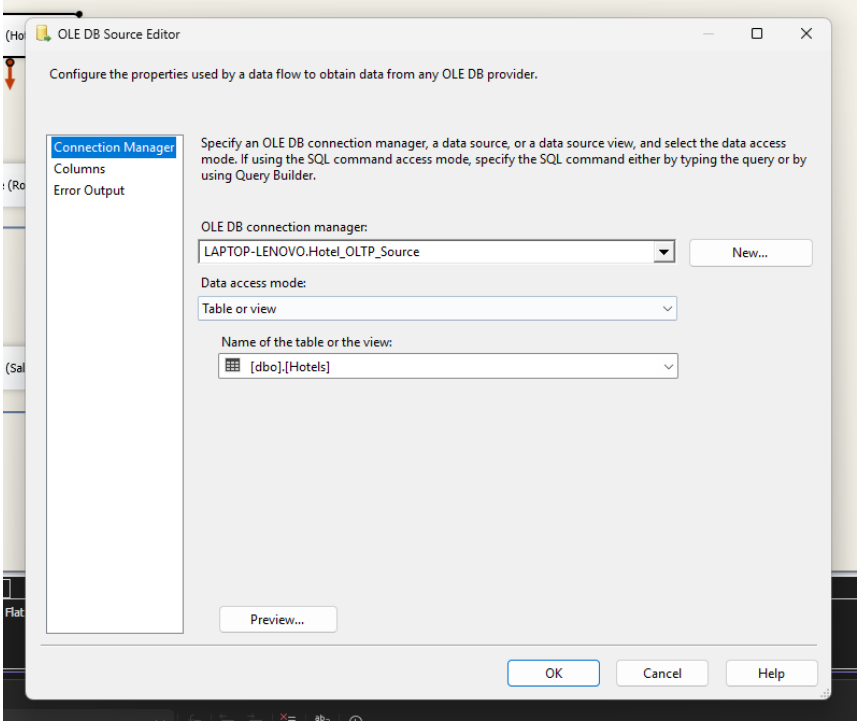
DBCC CHECKIDENT ('[Hotel_DW].[dbo].[DimSalesperson]', RESEED, 0);

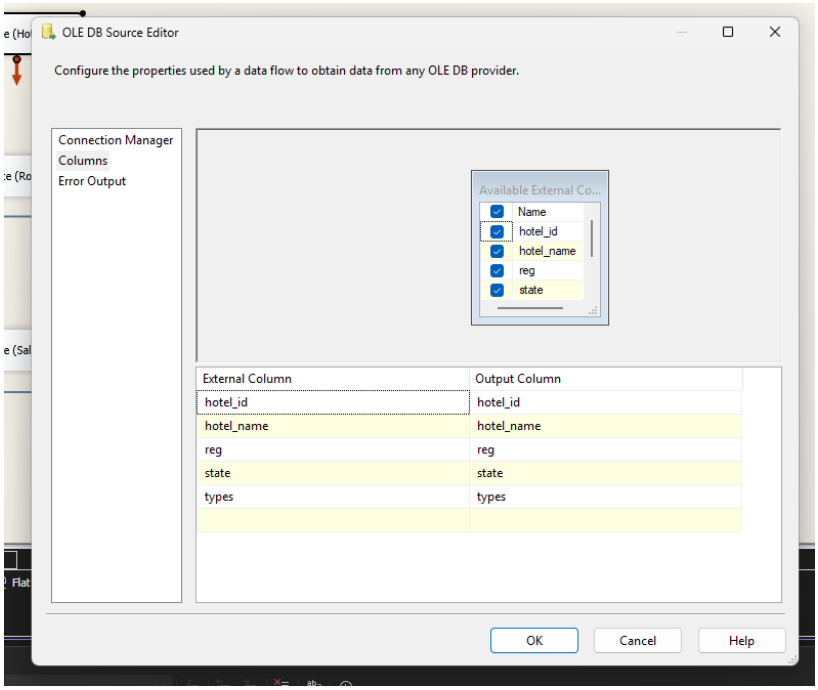


2. **DFT - Load SQL Dimensions:** Extracts metadata from the relational SQL source (Hotels, Rooms, Salesperson).



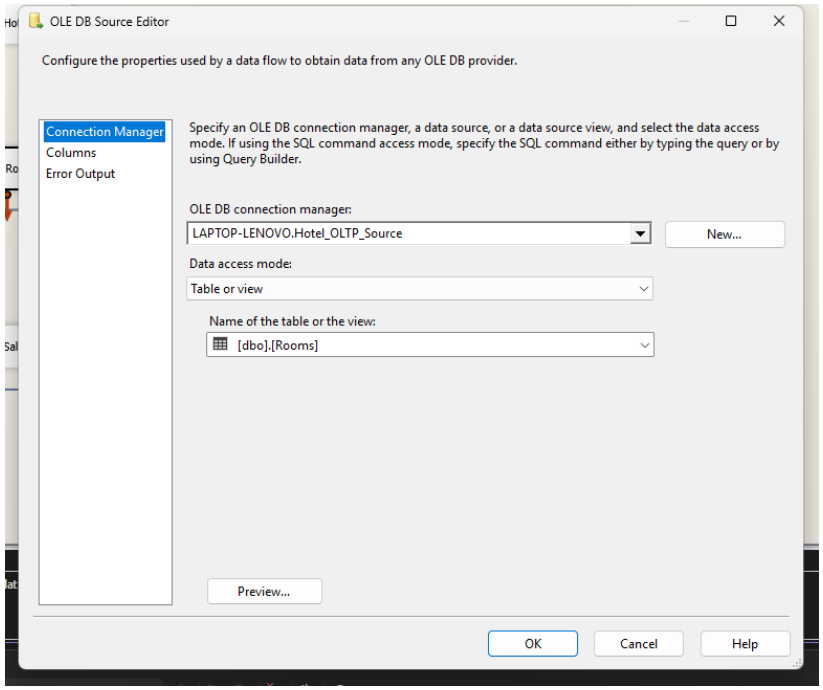
Hotels: DimHotel

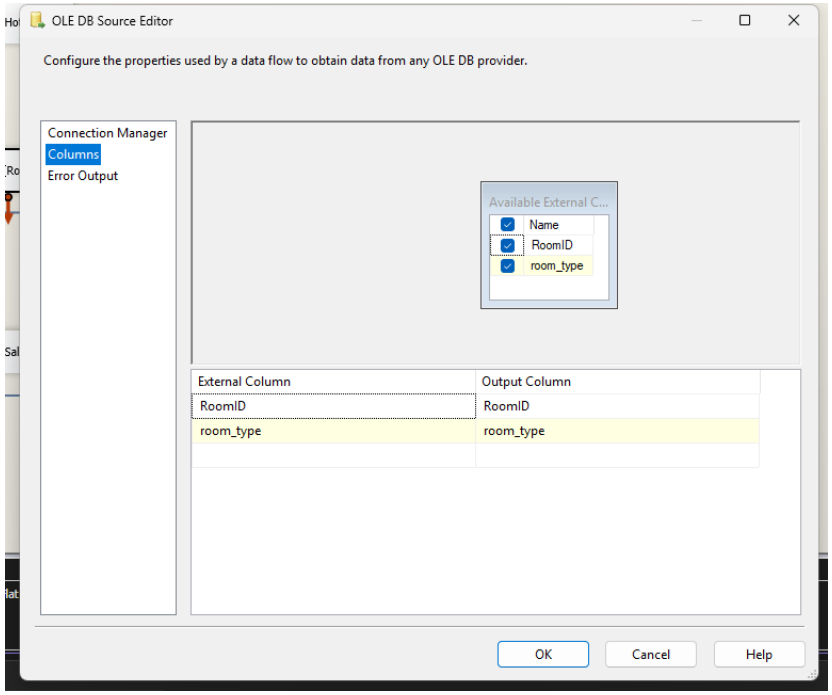




	HotelKey	hotel_id	hotel_name	reg	state	types
1	1	1	Le Meridien Sabah	East	Sabah	Resort Hotel
2	2	2	The Hilton Kuala Lumpur	Central	Kuala Lumpur	City Hotel
3	3	3	Lexis Suites Penang	North	Penang	City Hotel

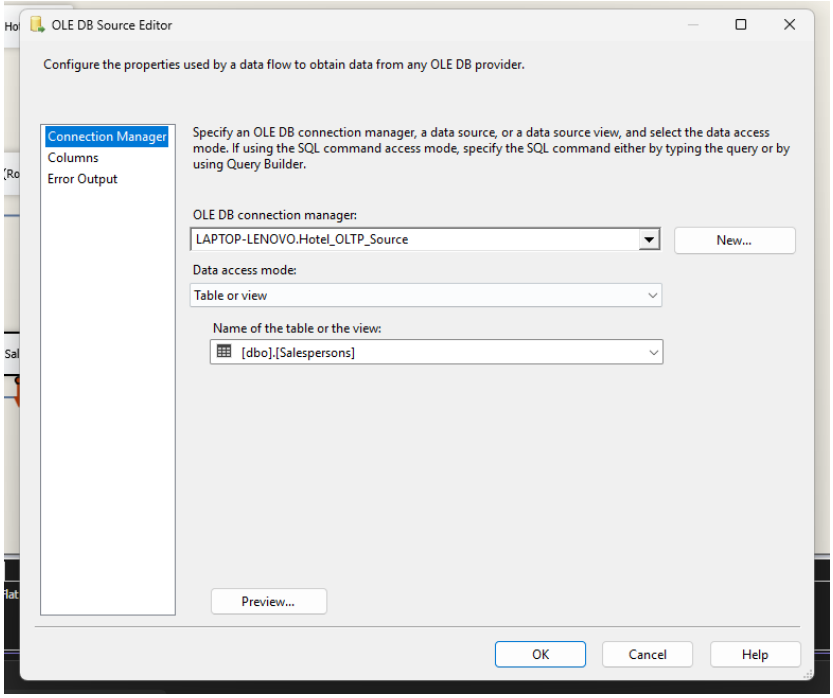
Rooms: DimRoom

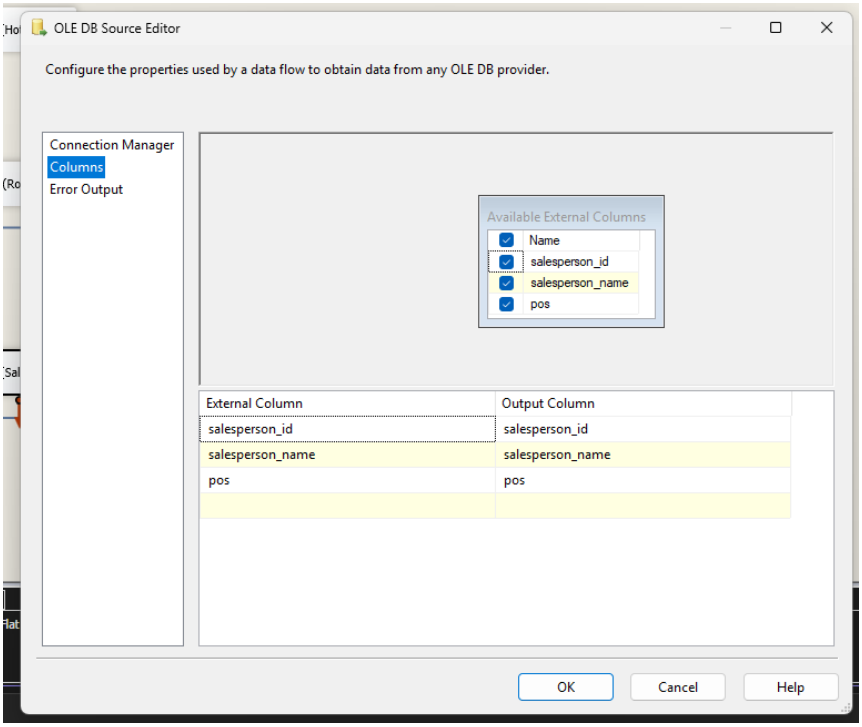




	RoomKey	RoomID	room_type
1	1	1	Deluxe Room
2	2	2	Single Room
3	3	3	Studio
4	4	4	Royal Suite

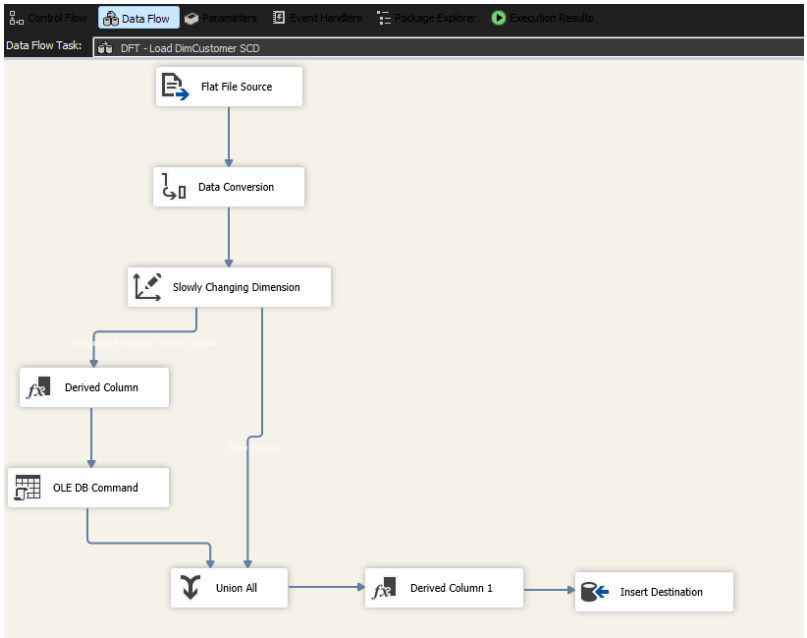
Salesperson: DimSalesperson



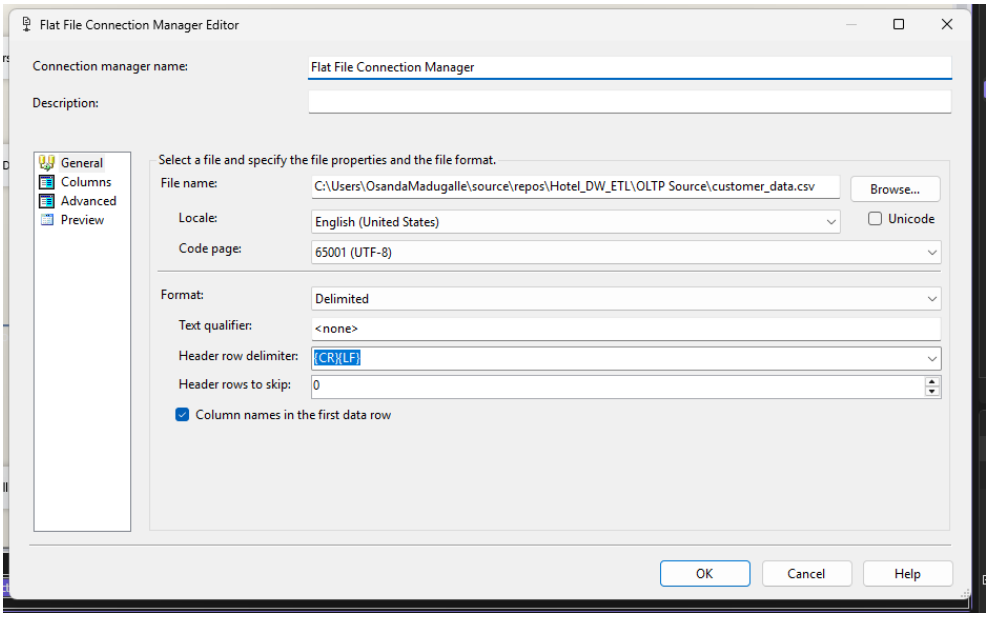


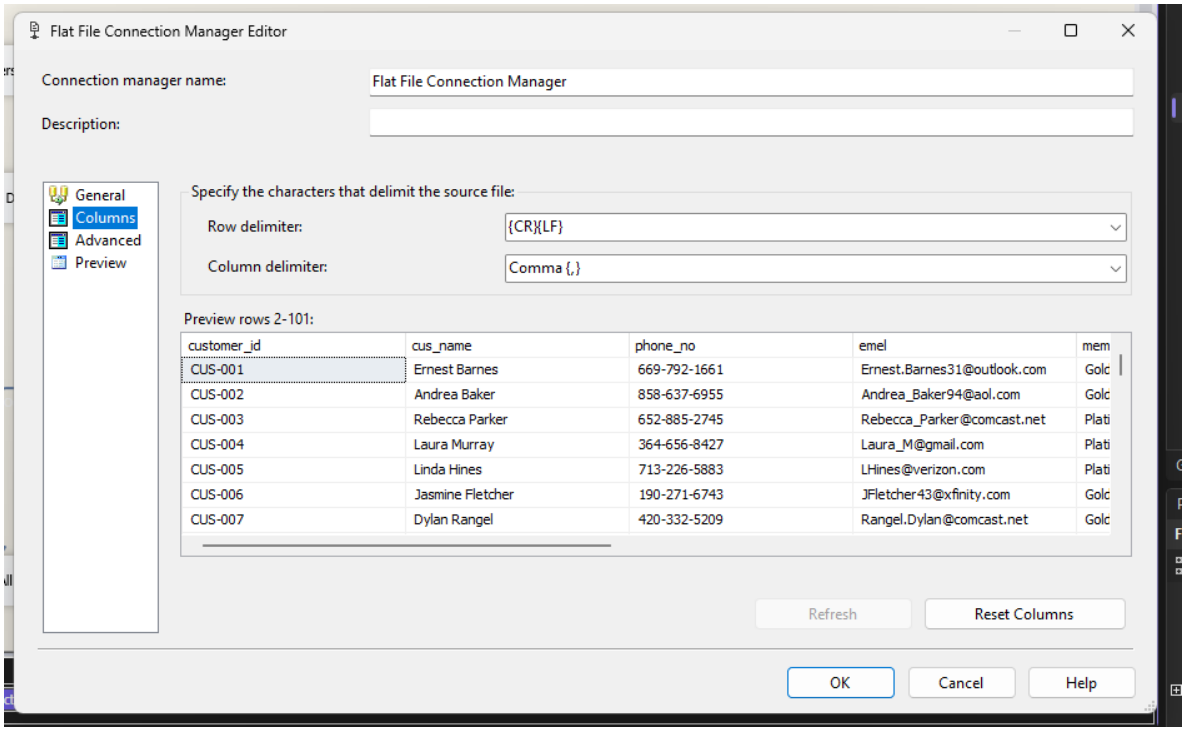
	SalespersonKey	salesperson_id	salesperson_name	pos
1	1	SP-001	Kelly	Sales Executive
2	2	SP-002	Amy	Sales Executive
3	3	SP-003	Derek	Sales Executive
4	4	SP-004	Kelly	Sales Manager
5	5	SP-005	Leo	Sales Manager
6	6	SP-006	Bernice	Sales Executive

3. **DFT - Load DimCustomer SCD:** Processes the CSV source and applies Slowly Changing Dimension logic.

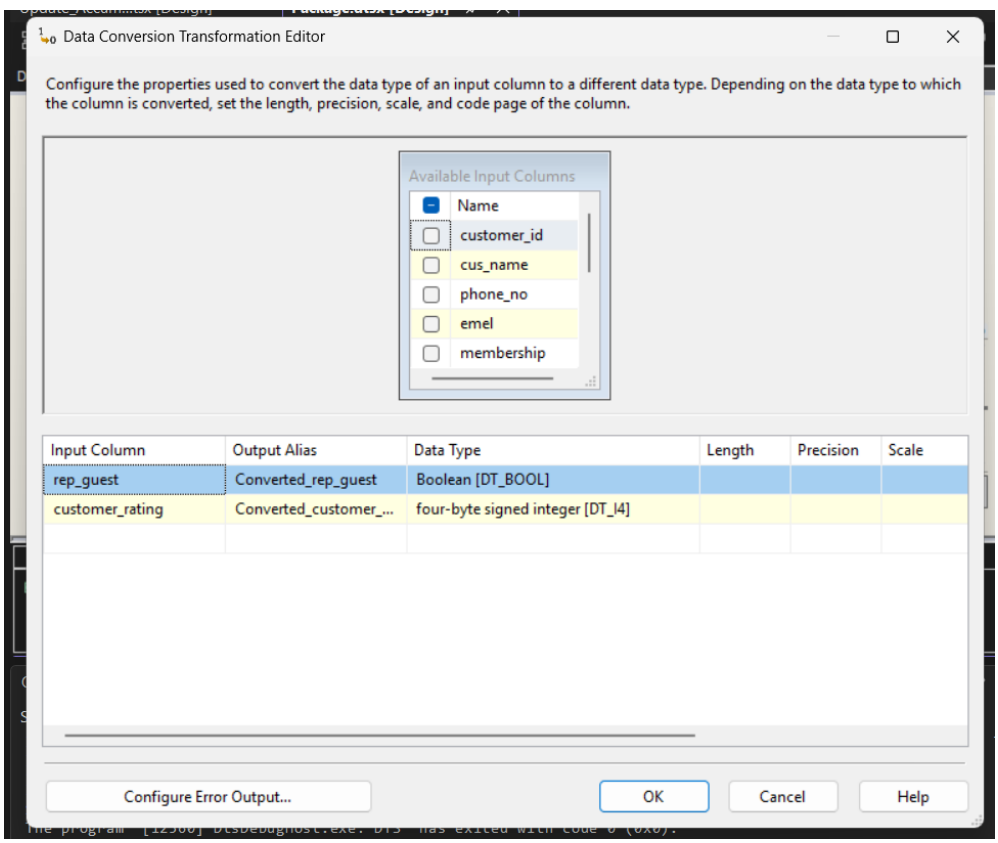


Flat File Source

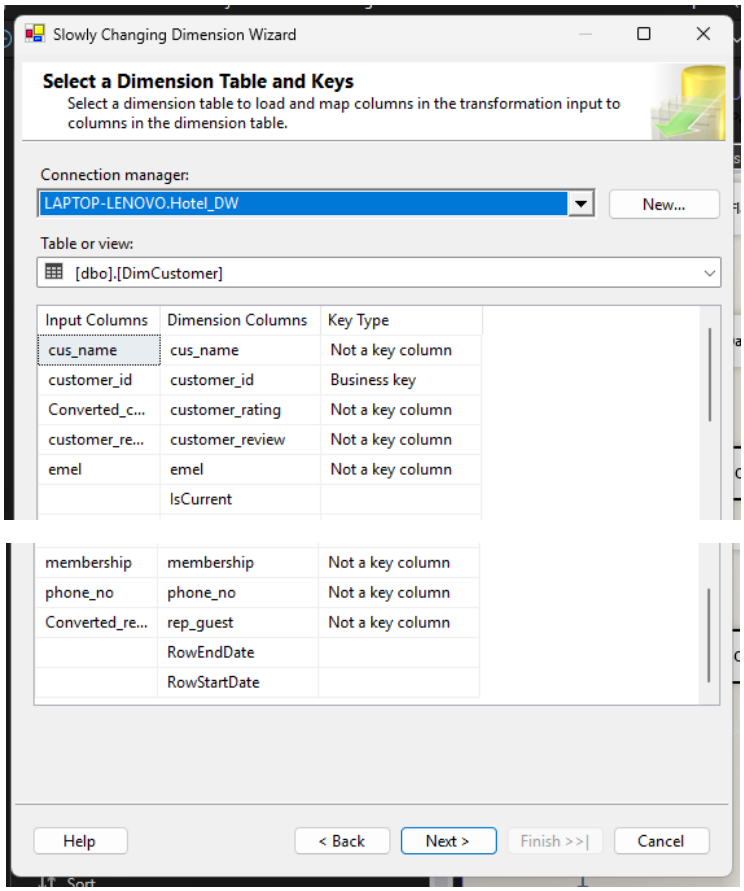


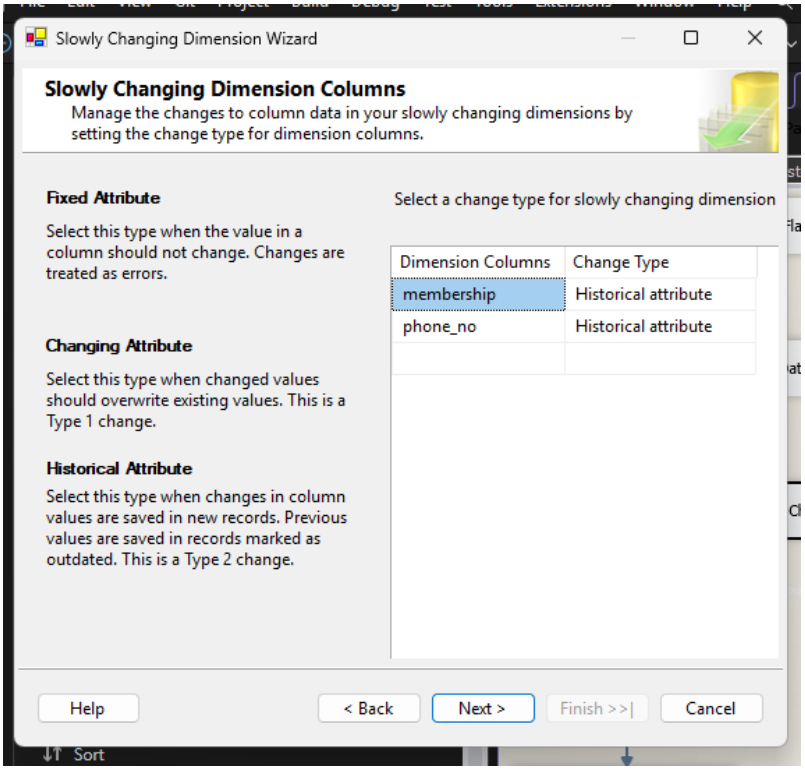


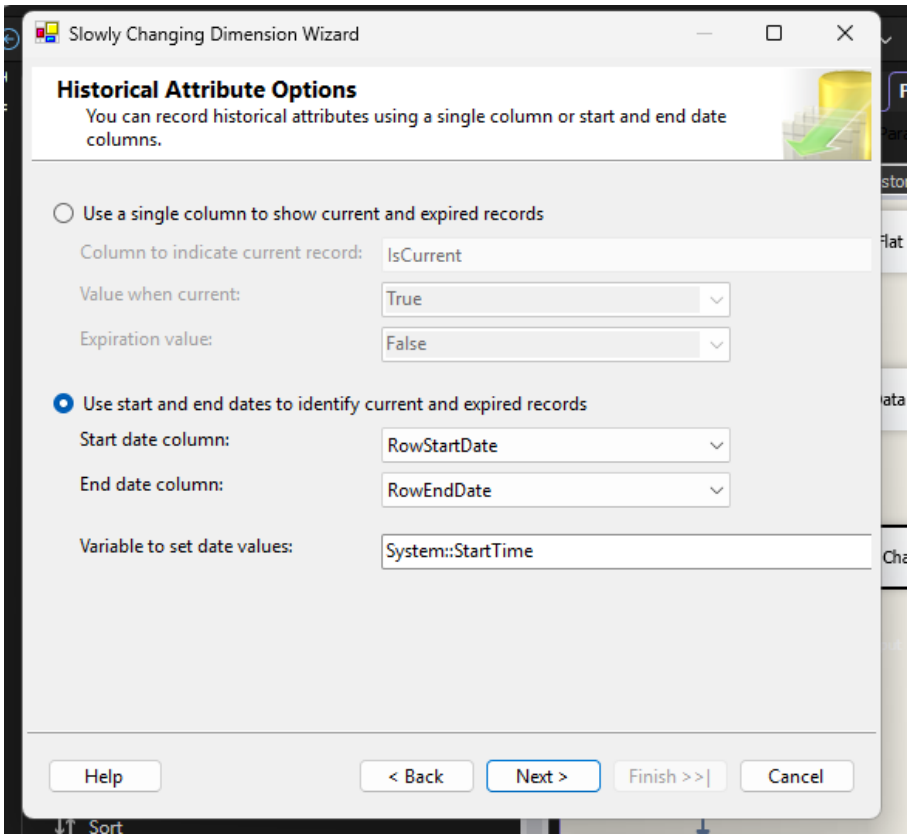
Data Conversion

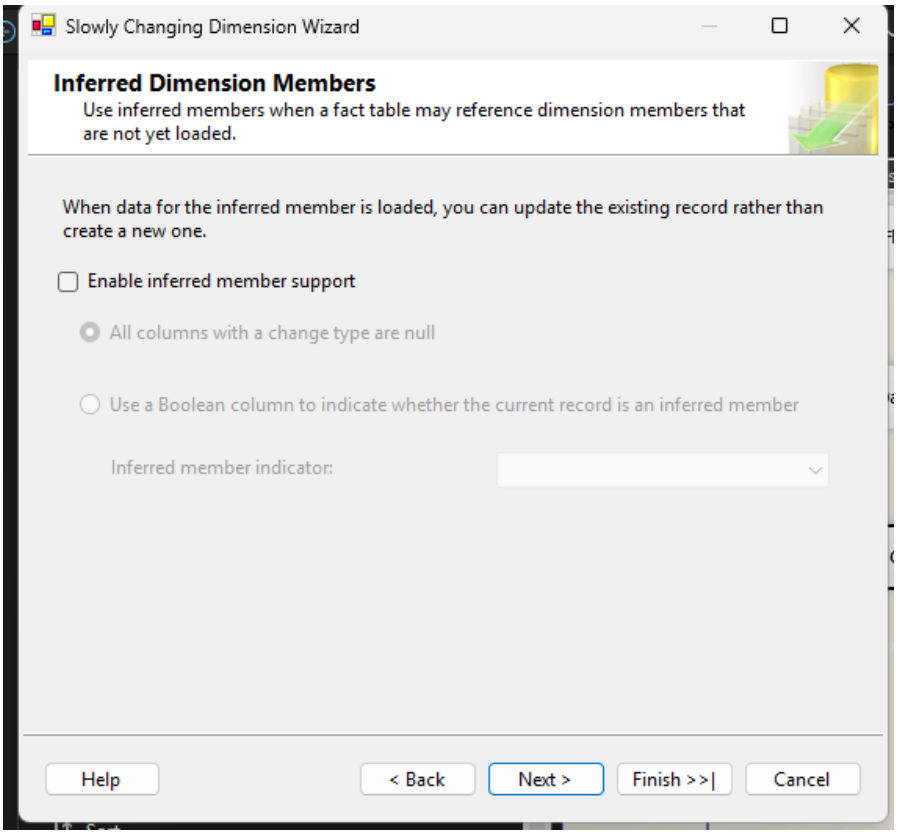


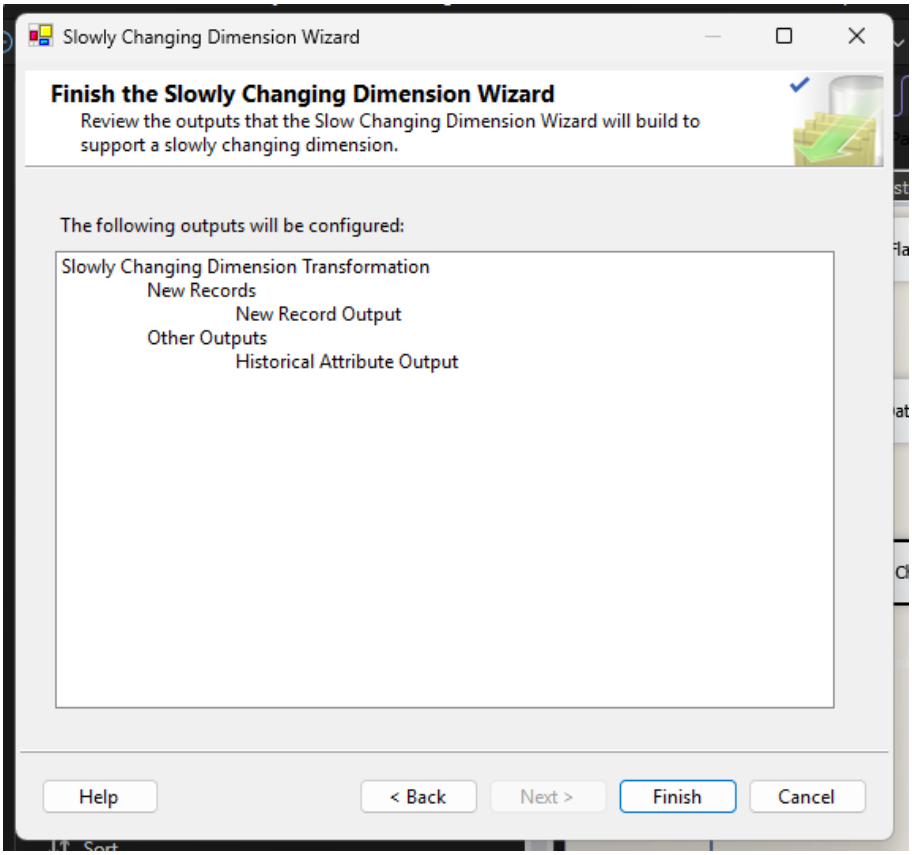
Slowly Changing Dimension



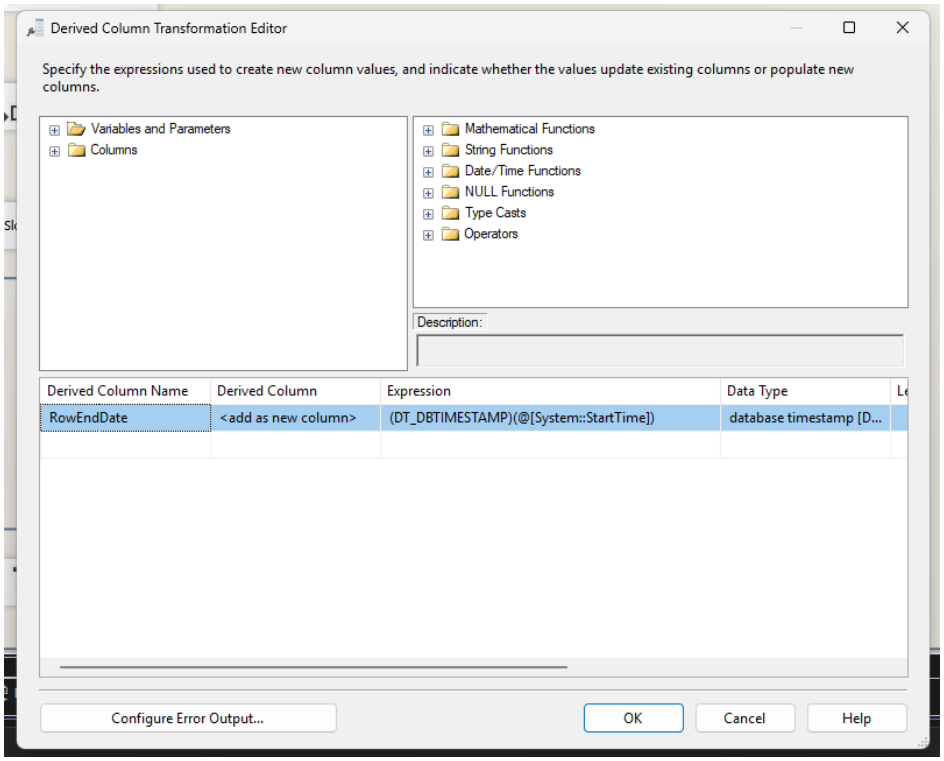



Slowly Changing Dimension Wizard
Historical Attribute Options
You can record historical attributes using a single column or start and end date columns.
Use a single column to show current and expired records
Column to indicate current record:
IsCurrent
Value when current:
True
Expiration value:
False
Use start and end dates to identify current and expired records
Start date column:
RowStartDate
End date column:
RowEndDate
Variable to set date values:
System::StartTime
Help
< Back
Next >
Finish >>|
Cancel

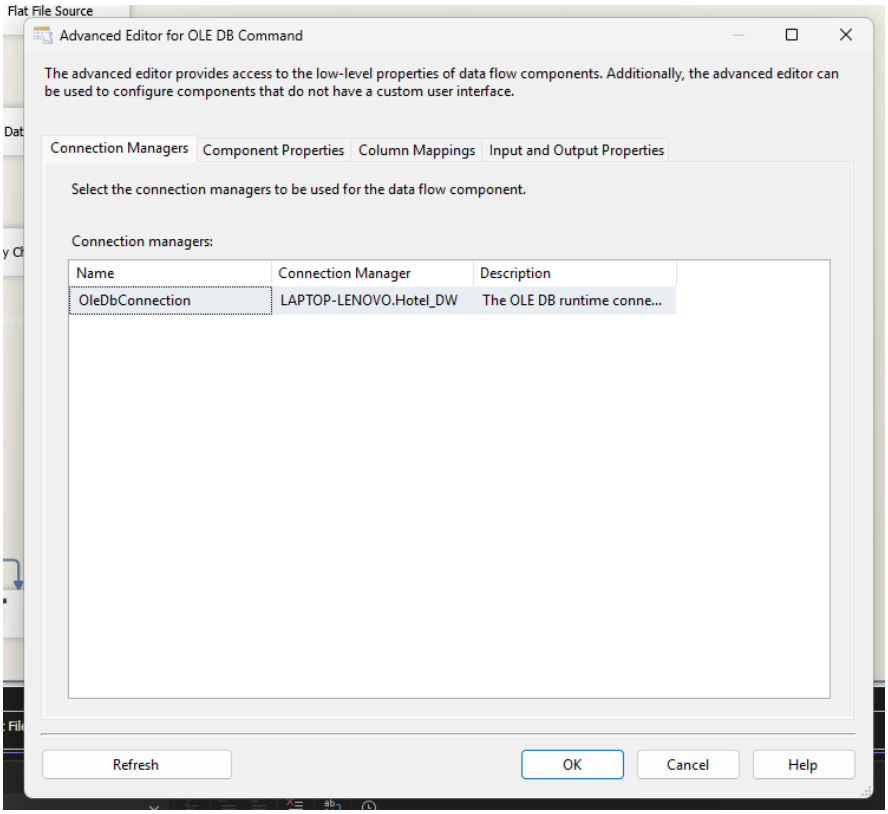

Slowly Changing Dimension Wizard
Inferred Dimension Members
Use inferred members when a fact table may reference dimension members that are not yet loaded.
When data for the inferred member is loaded, you can update the existing record rather than create a new one.
Enable inferred member support
All columns with a change type are null
Use a Boolean column to indicate whether the current record is an inferred member
Inferred member indicator:
Help
< Back
Next >
Finish >>|
Cancel

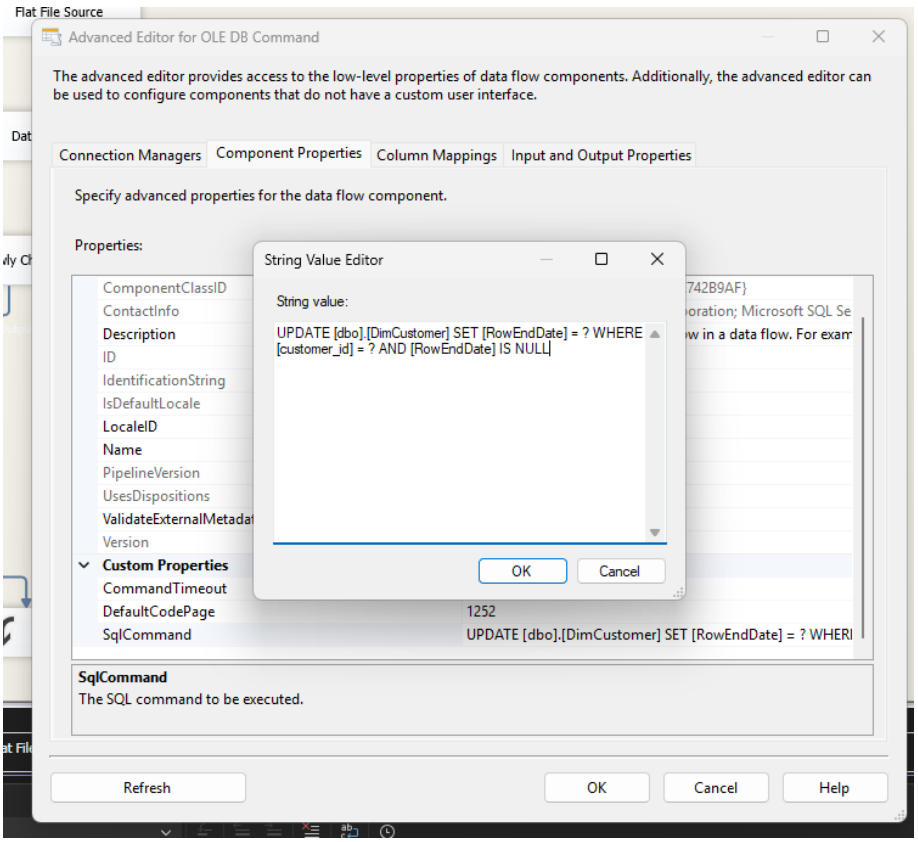


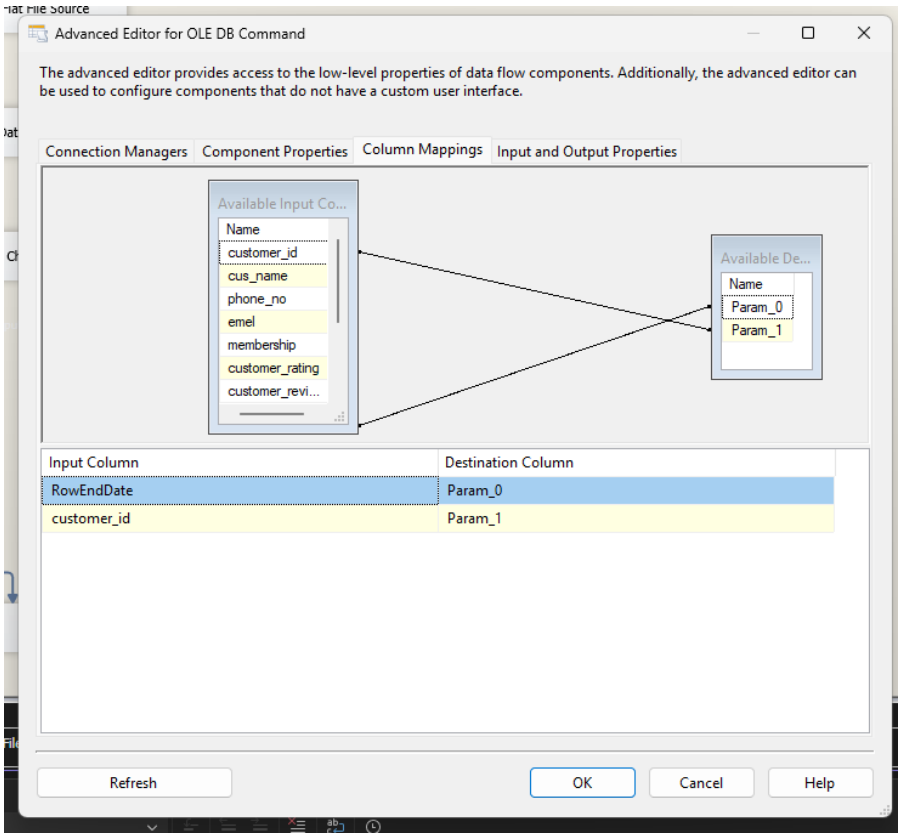
Derived Column



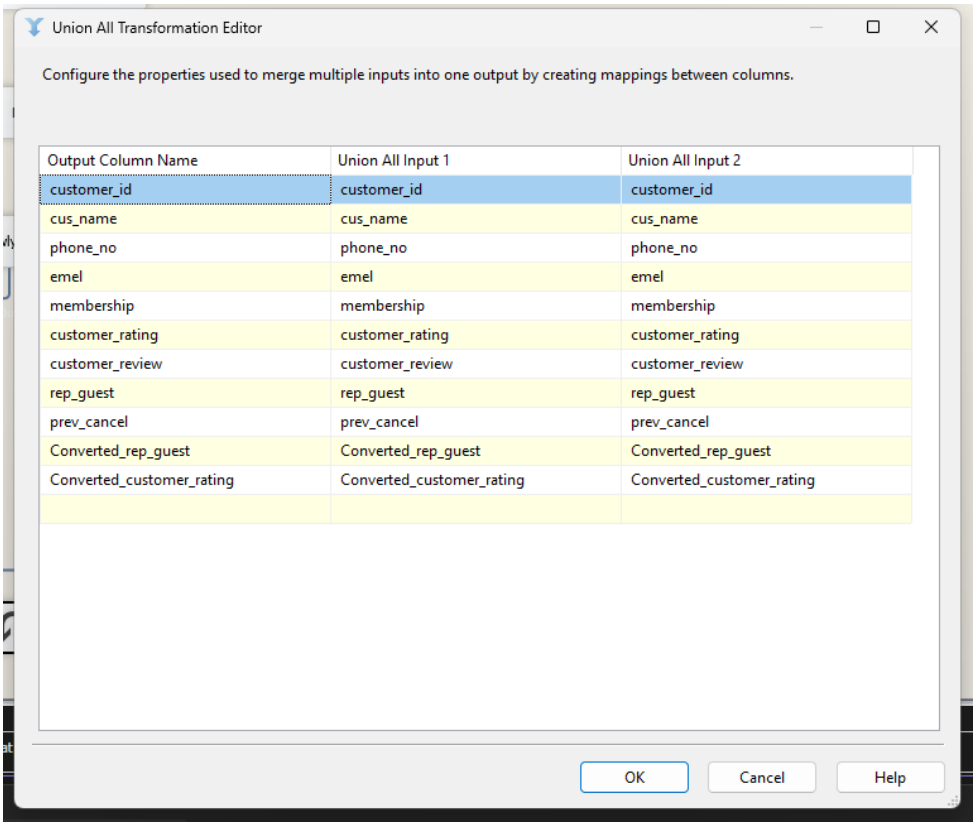
OLE DB Command



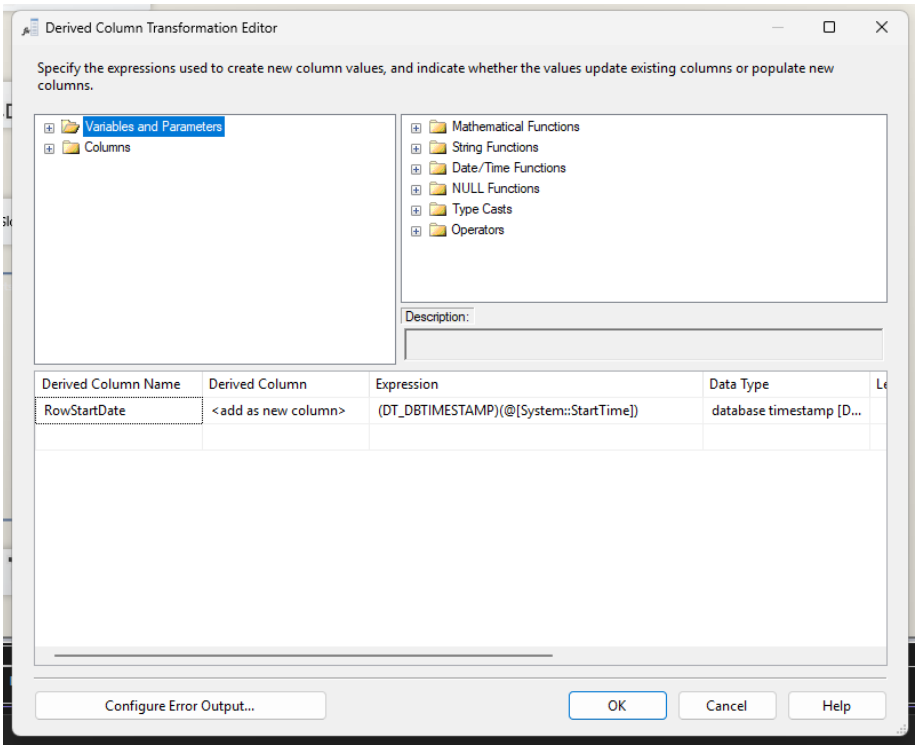



Advanced Editor for OLE DB Command
The advanced editor provides access to the low-level properties of data flow components. Additionally, the advanced editor can be used to configure components that do not have a custom user interface.
Connection Managers
Component Properties
Column Mappings
Input and Output Properties
Available Input Co...
Name
customer_id
cus_name
phone_no
emel
membership
customer_rating
customer_revi...
Available De...
Name
Param_0
Param_1
Input Column
Destination Column
RowEndDate
Param_0
customer_id
Param_1
Refresh
OK
Cancel
Help

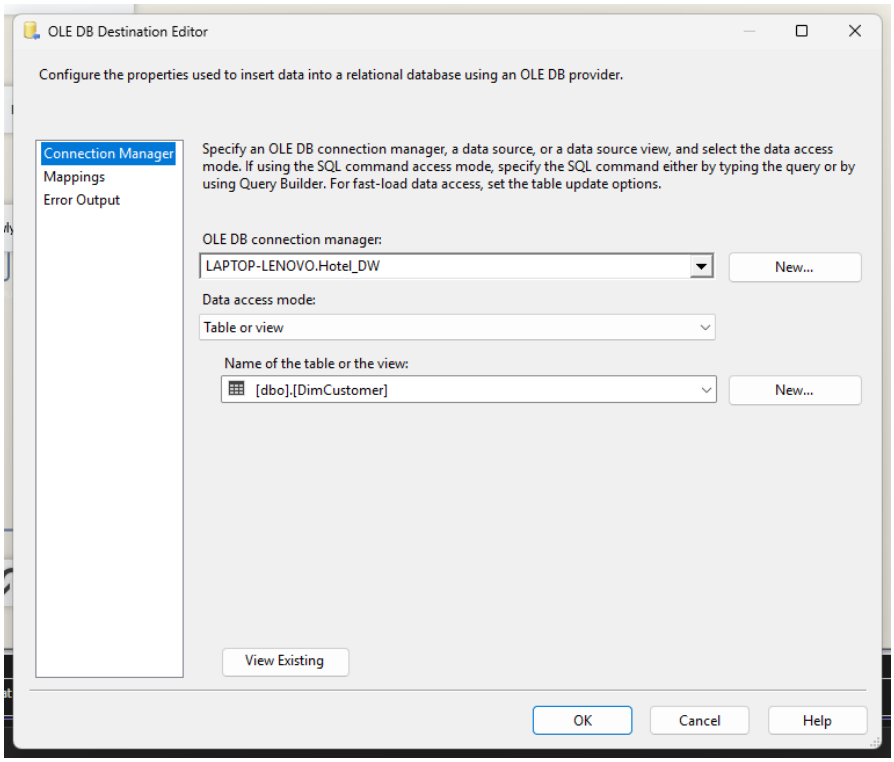
Union All

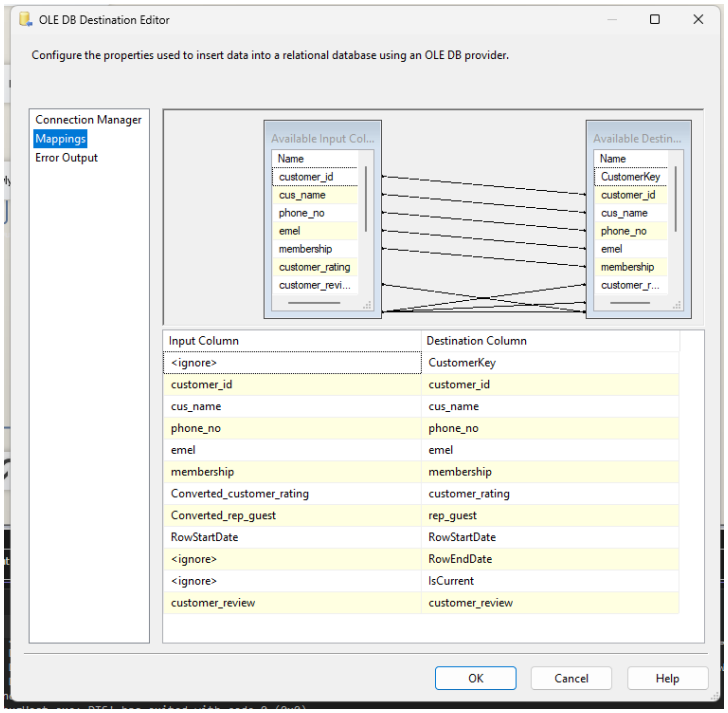

Union All Transformation Editor
Configure the properties used to merge multiple inputs into one output by creating mappings between columns.
Output Column Name
Union All Input 1
Union All Input 2
customer_id
customer_id
customer_id
cus_name
cus_name
cus_name
phone_no
phone_no
phone_no
emel
emel
emel
membership
membership
membership
customer_rating
customer_rating
customer_rating
customer_review
customer_review
customer_review
rep_guest
rep_guest
rep_guest
prev_cancel
prev_cancel
prev_cancel
Converted_rep_guest
Converted_rep_guest
Converted_rep_guest
Converted_customer_rating
Converted_customer_rating
Converted_customer_rating
OK
Cancel
Help

Derived Column 1



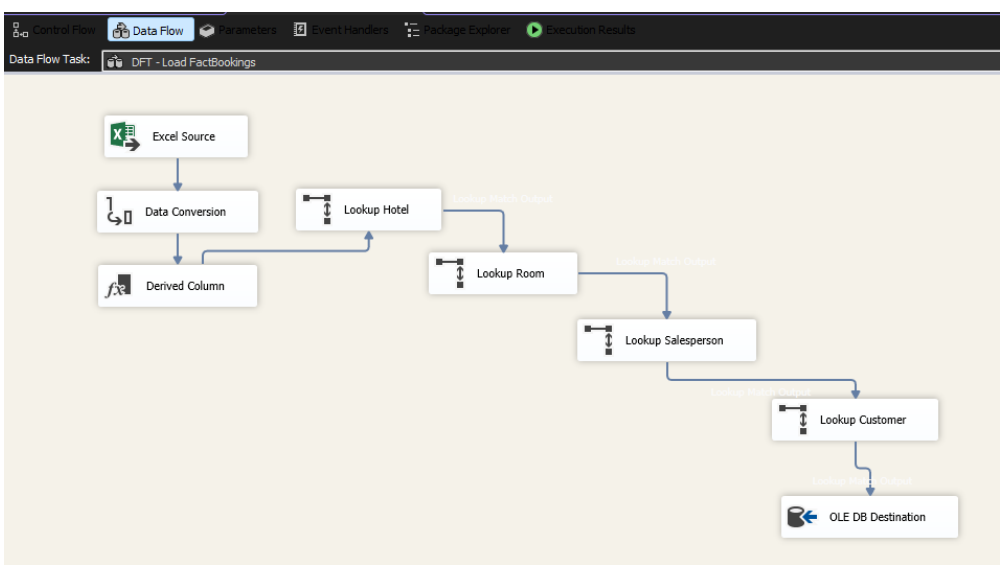
Insert Destination

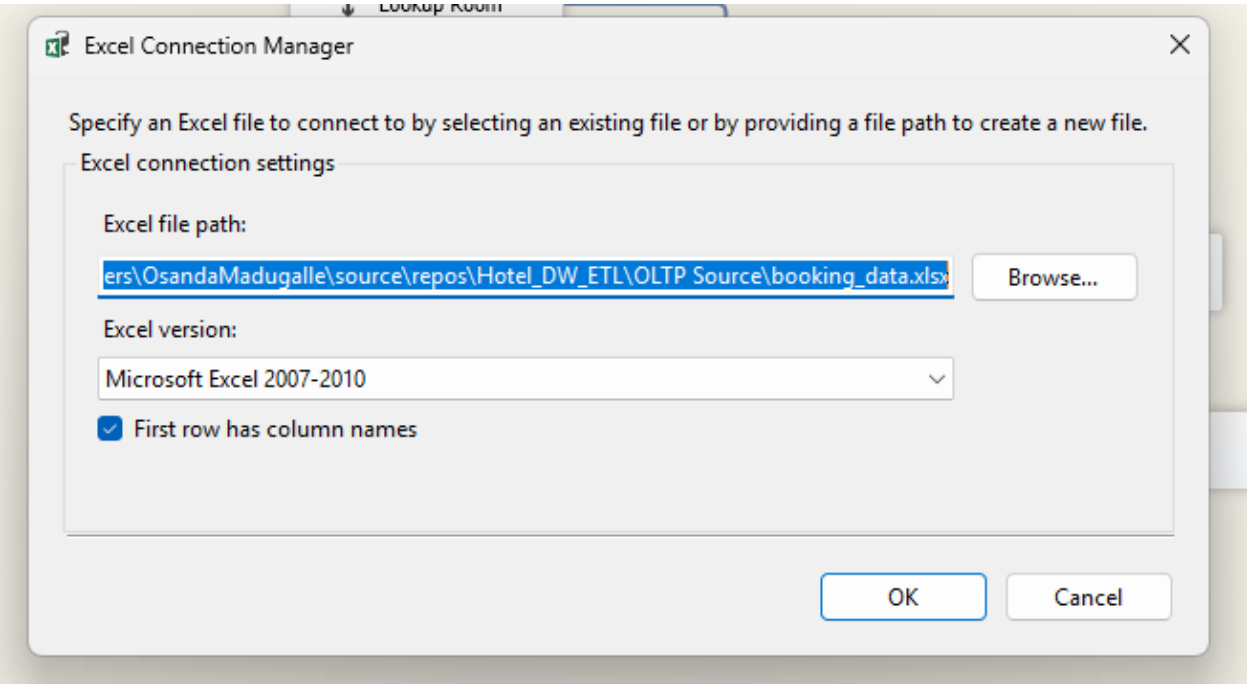


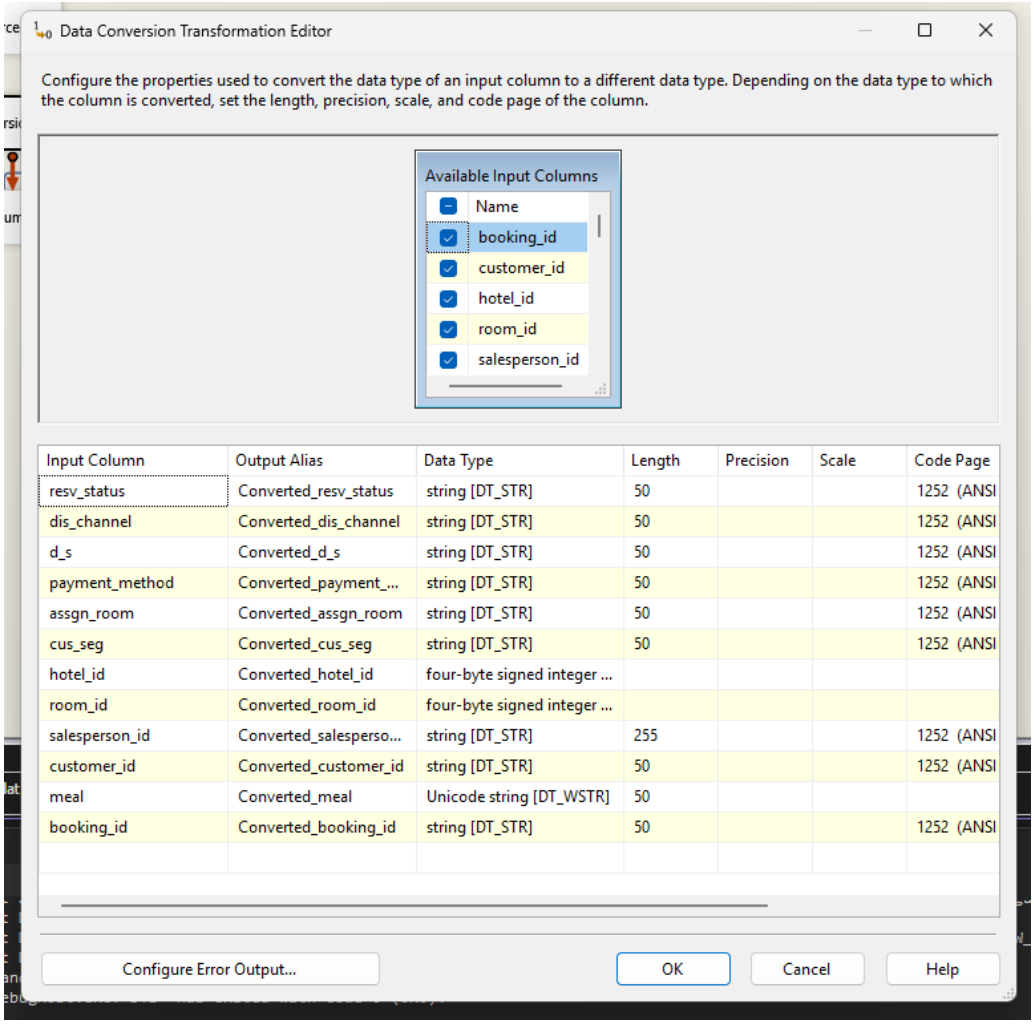


	CustomerKey	customer_id	cus_name	phone_no	emel	membership	customer_rating	rep_guest	RowStartDate
1	1	CUS-001	Ernest Barnes	669-792-1661	Ernest.Barnes31@outlook.com	Gold	5	0	2026-03-30 21:03:45.00
2	2	CUS-002	Andrea Baker	858-637-6955	Andrea_Baker94@aol.com	Gold	4	0	2026-03-30 21:03:45.00
3	3	CUS-003	Rebecca Parker	652-885-2745	Rebecca_Parker@comcast.net	Platinum	5	0	2026-03-30 21:03:45.00
4	4	CUS-004	Laura Murray	364-656-8427	Laura_M@gmail.com	Platinum	4	0	2026-03-30 21:03:45.00
5	5	CUS-005	Linda Hines	713-226-5883	LHines@verizon.com	Platinum	5	0	2026-03-30 21:03:45.00
6	6	CUS-006	Jasmine Fletcher	190-271-6743	JFletcher43@xfinity.com	Gold	5	0	2026-03-30 21:03:45.00
7	7	CUS-007	Dylan Rangel	420-332-5209	Rangel.Dylan@comcast.net	Gold	5	0	2026-03-30 21:03:45.00
8	8	CUS-008	William Velez	286-669-4333	Velez_William@mail.com	Gold	4	0	2026-03-30 21:03:45.00
9	9	CUS-009	Steven Murphy	341-726-5787	Steven.Murphy54@aol.com	Gold	5	0	2026-03-30 21:03:45.00
10	10	CUS-010	Michael Moore	316-648-6176	MichaelMoore81@outlook.com	Gold	4	0	2026-03-30 21:03:45.00
11	11	CUS-011	Priscilla Collins PhD	833-887-7898	PhD.Priscilla74@att.com	Platinum	5	0	2026-03-30 21:03:45.00

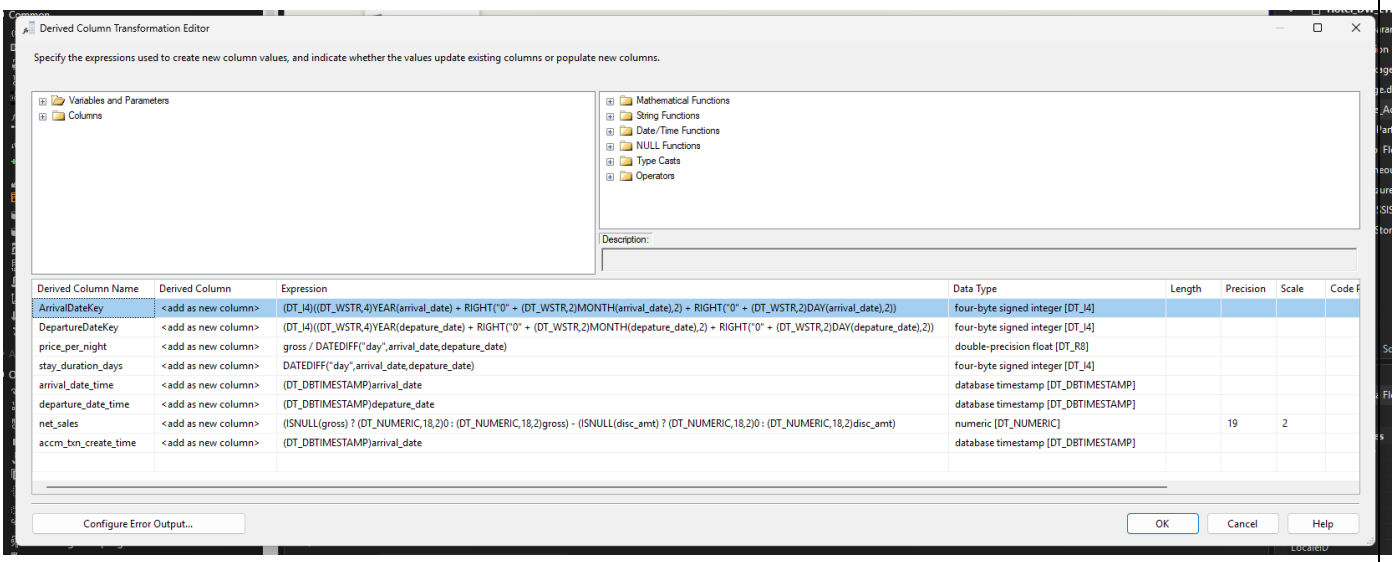
4. **DFT - Load FactBookings:** The final process that aggregates all sources and links them to the prepared dimensions.



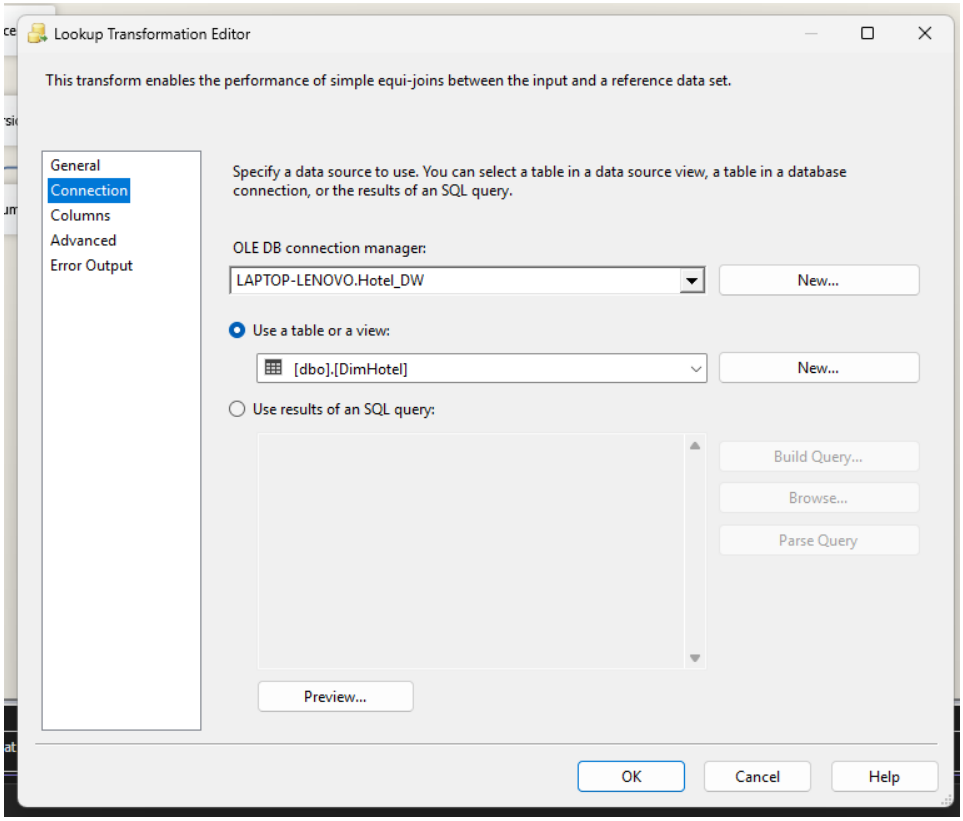


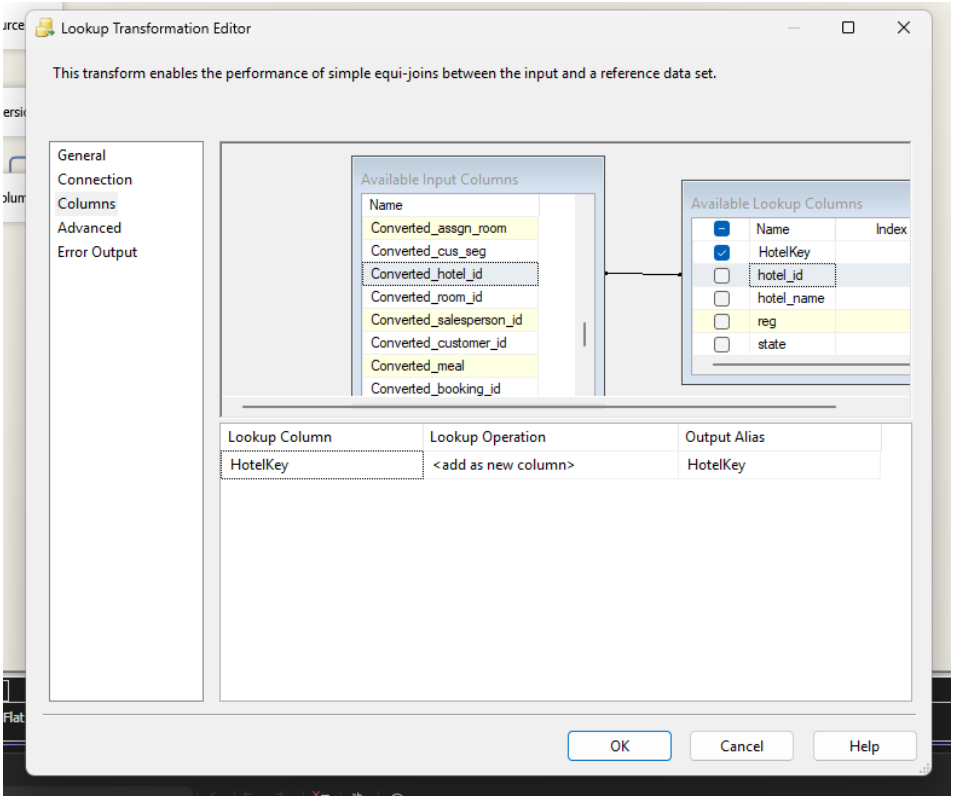


Input Column	Output Alias	Data Type	Length	Precision	Scale	Code Page
resv_status	Converted_resv_status	string [DT_STR]	50			1252 (ANSI
dis_channel	Converted_dis_channel	string [DT_STR]	50			1252 (ANSI
d_s	Converted_d_s	string [DT_STR]	50			1252 (ANSI
payment_method	Converted_payment_...	string [DT_STR]	50			1252 (ANSI
assgn_room	Converted_assgn_room	string [DT_STR]	50			1252 (ANSI
cus_seg	Converted_cus_seg	string [DT_STR]	50			1252 (ANSI
hotel_id	Converted_hotel_id	four-byte signed integer ...				
room_id	Converted_room_id	four-byte signed integer ...				
salesperson_id	Converted_salesperso...	string [DT_STR]	255			1252 (ANSI
customer_id	Converted_customer_id	string [DT_STR]	50			1252 (ANSI
meal	Converted_meal	Unicode string [DT_WSTR]	50			
booking_id	Converted_booking_id	string [DT_STR]	50			1252 (ANSI

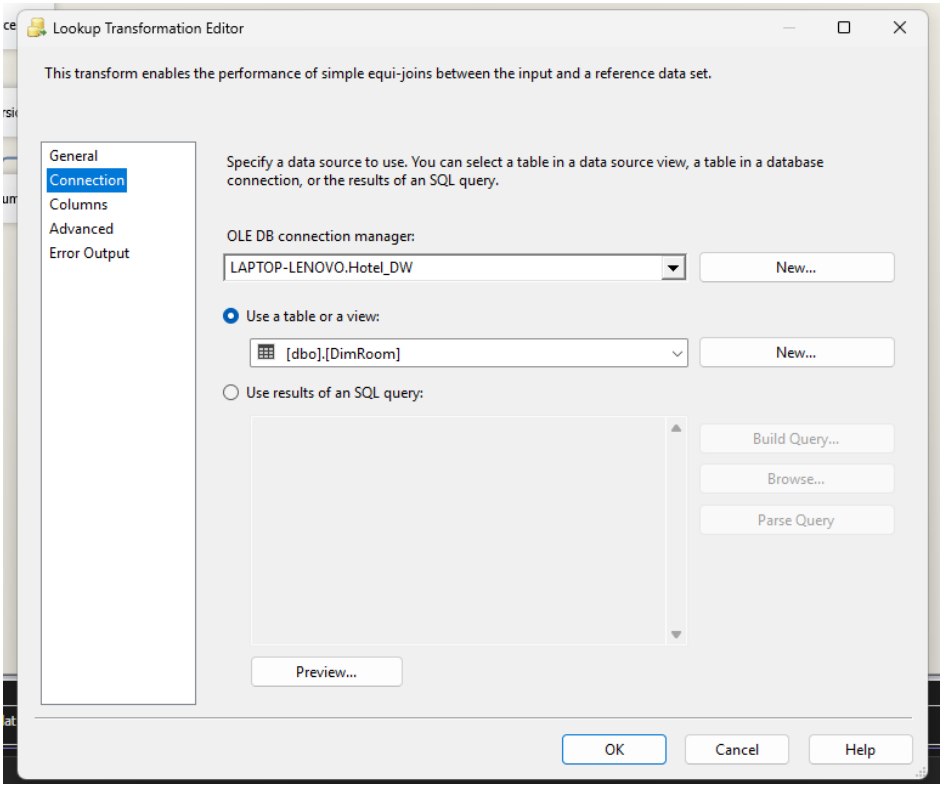


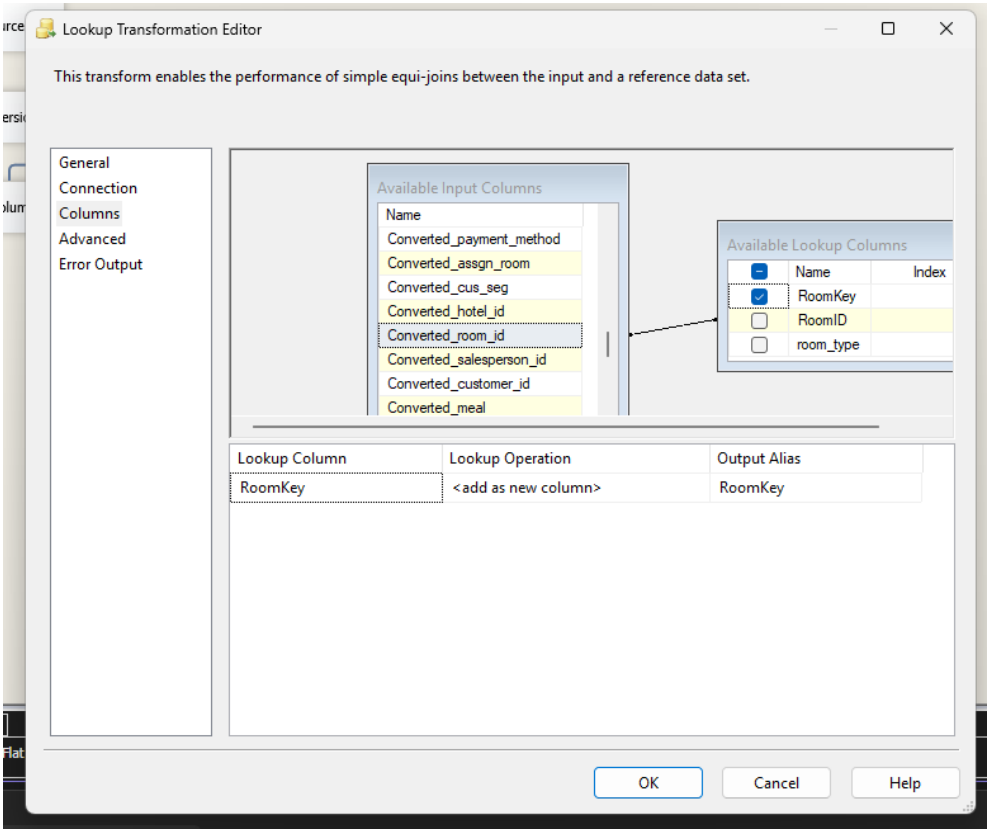
Lookup Hotel



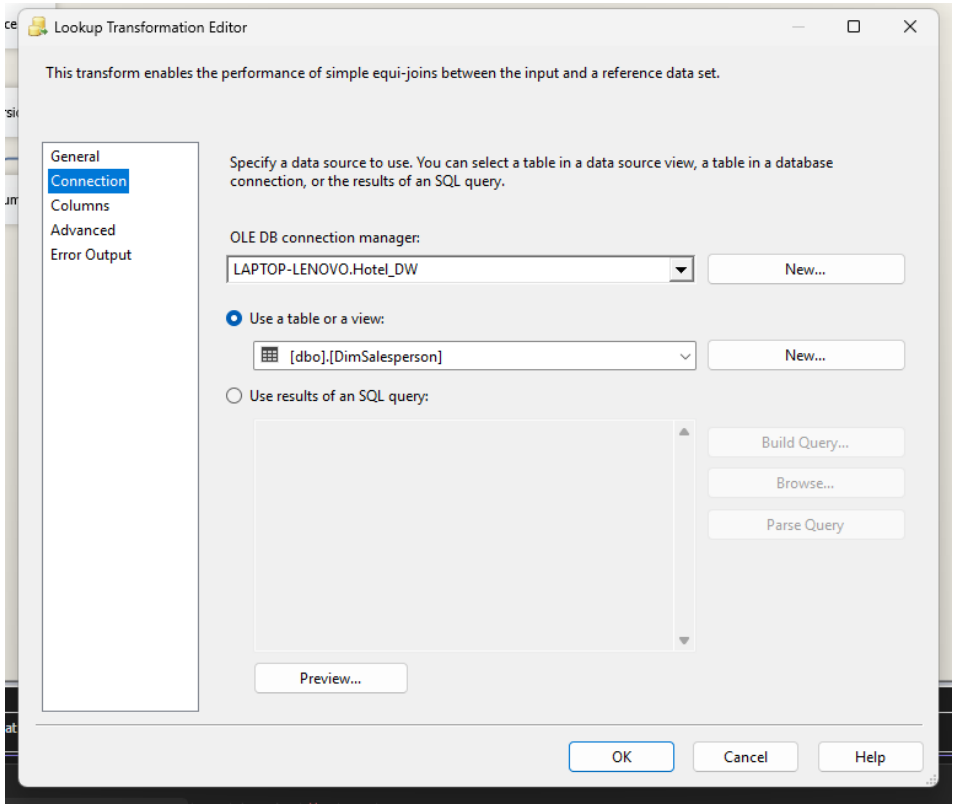

Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Available Input Columns
Name
Converted_assgn_room
Converted_cus_seg
Converted_hotel_id
Converted_room_id
Converted_salesperson_id
Converted_customer_id
Converted_meal
Converted_booking_id
Available Lookup Columns
Name
Index
HotelKey
hotel_id
hotel_name
reg
state
Lookup Column
Lookup Operation
Output Alias
HotelKey
<add as new column>
HotelKey
OK
Cancel
Help

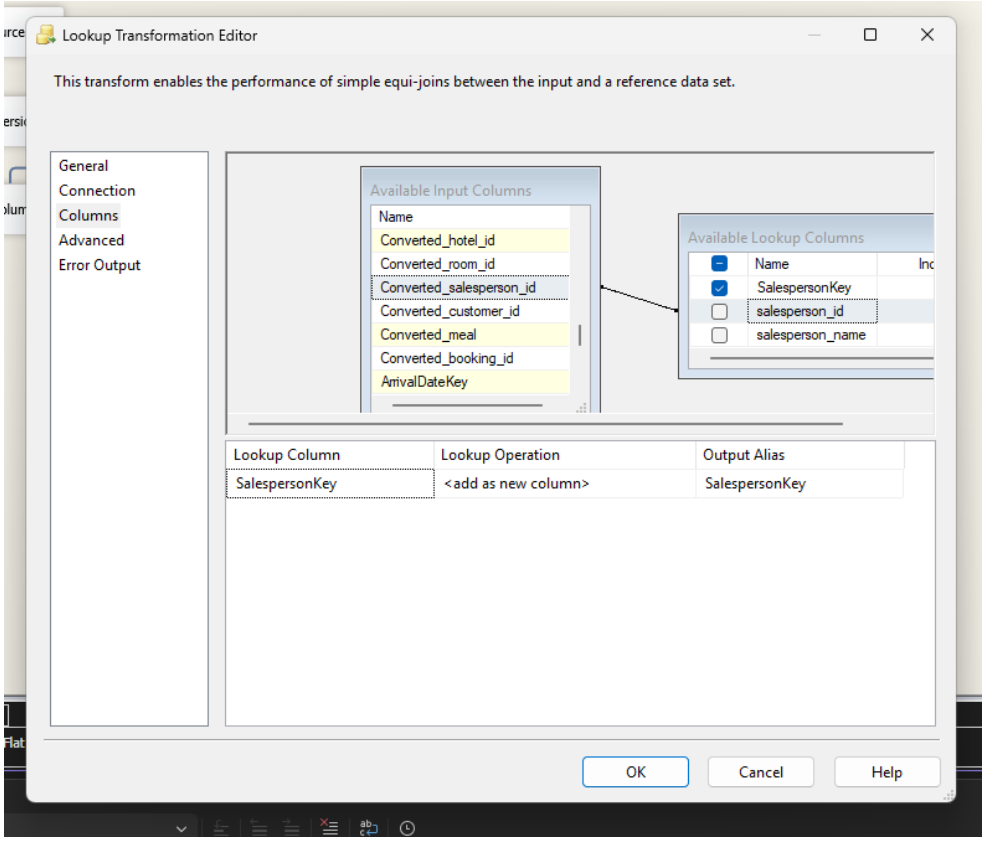
Lookup Room


Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.
OLE DB connection manager:
LAPTOP-LENOVO.Hotel_DW
New...
Use a table or a view:
[dbo].[DimRoom]
New...
Use results of an SQL query:
Build Query...
Browse...
Parse Query
Preview...
OK
Cancel
Help

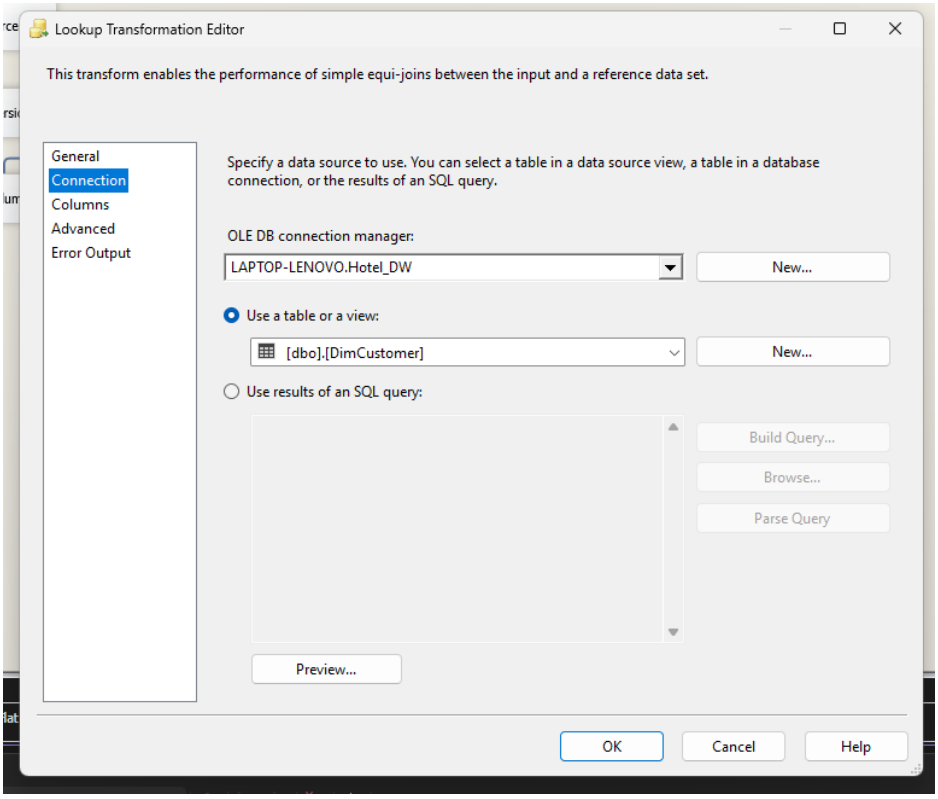


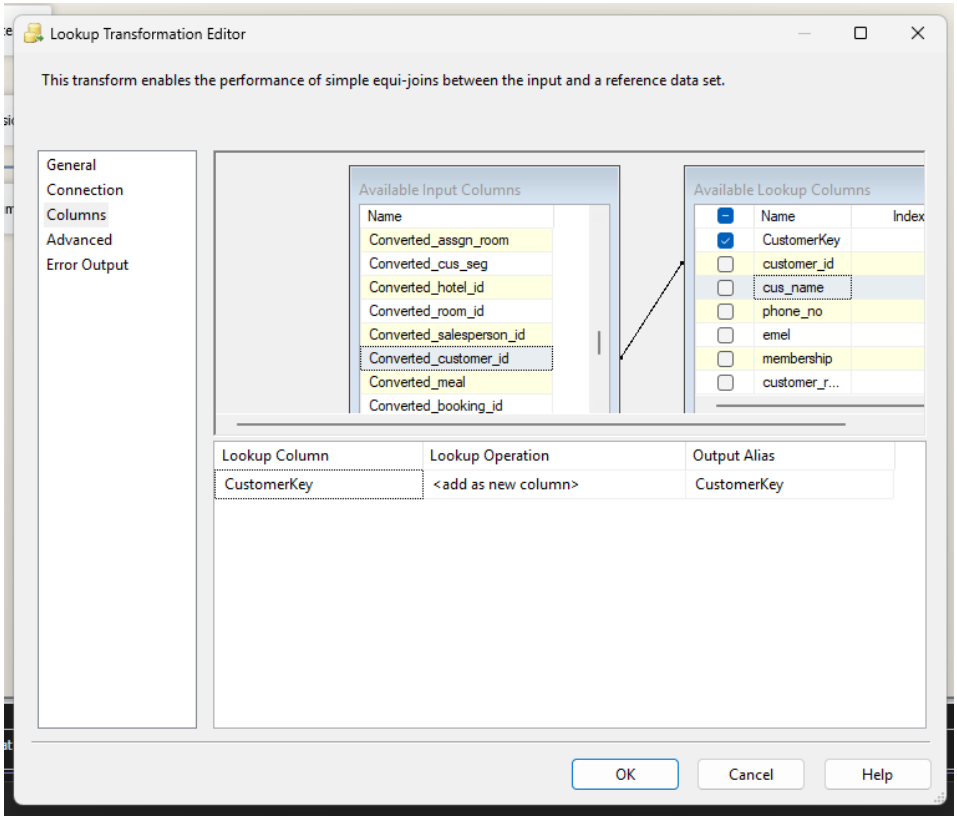
Lookup Salesperson

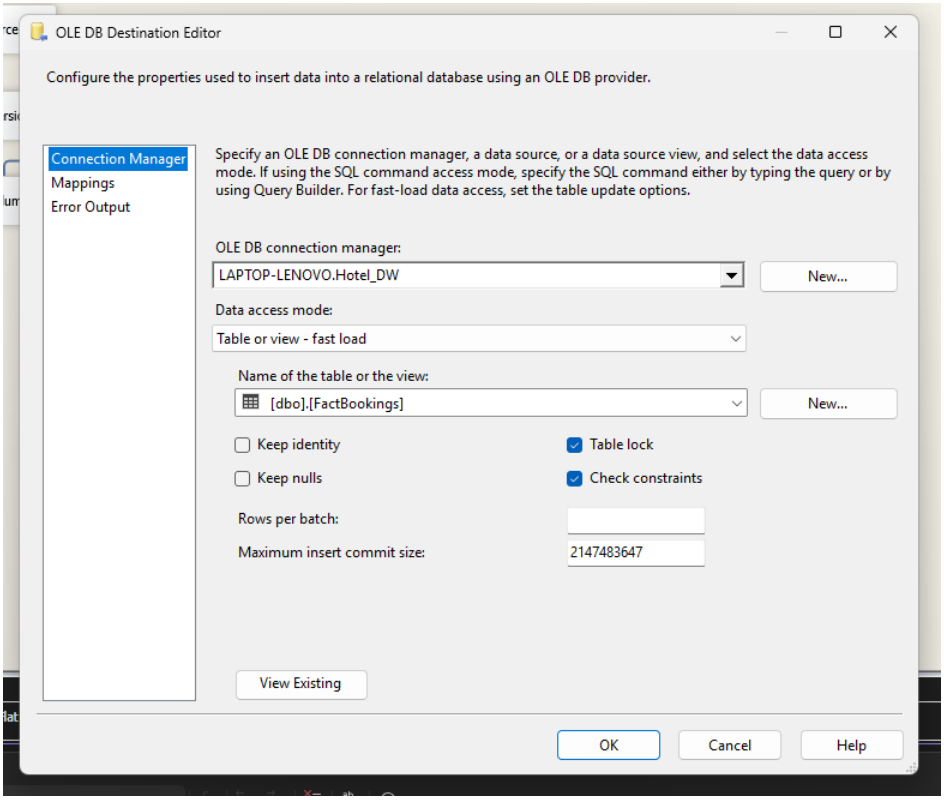




Lookup Customer




Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Available Input Columns
Name
Converted_assgn_room
Converted_cus_seg
Converted_hotel_id
Converted_room_id
Converted_salesperson_id
Converted_customer_id
Converted_meal
Converted_booking_id
Available Lookup Columns
Name
Index
CustomerKey
customer_id
cus_name
phone_no
emel
membership
customer_r...
Lookup Column
Lookup Operation
Output Alias
CustomerKey
<add as new column>
CustomerKey
OK
Cancel
Help


OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.
OLE DB connection manager:
LAPTOP-LENOVO.Hotel_DW
New...
Data access mode:
Table or view - fast load
Name of the table or the view:
[dbo].[FactBookings]
New...
Keep identity
Table lock
Keep nulls
Check constraints
Rows per batch:
Maximum insert commit size:
2147483647
View Existing
OK
Cancel
Help

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager
Mappings
Error Output

Available Input C...

Name
booking_id
customer_id
hotel_id
room_id
salesperson_id
resv_status
arrival_date

Available Destinati...

Name
BookingKey
HotelKey
RoomKey
SalespersonKey
CustomerKey
ArrivalDateKey
DepartureDat...

Input Column	Destination Column
<ignore>	BookingKey
HotelKey	HotelKey
RoomKey	RoomKey
SalespersonKey	SalespersonKey
CustomerKey	CustomerKey
ArrivalDateKey	ArrivalDateKey
DepartureDateKey	DepartureDateKey
Converted_resv_status	resv_status
nights	nights
adult	adult
child	child
Converted_dis_channel	dis_channel
Converted_payment_method	payment_method
Converted_cus_seg	cus_seg
package_	package_name
gross	gross_amount
disc_amt	discount_amt
net_sales	net_sales
ADR	adr
comm_pay	comm_pay
price_per_night	price_per_night
arrival_date_time	arrival_date_time
departure_date_time	departure_date_time
<ignore>	stay_duration_days
Converted_meal	meal
accm_txn_create_time	accm_txn_create_time
<ignore>	accm_txn_complete_time
<ignore>	txn_process_time_hours
Converted_booking_id	BookingID

OK Cancel Help

5.2 Implementation of Slowly Changing Dimensions (SCD)

We implemented **SCD Type 2** for the DimCustomer table. This is critical for the hospitality industry to track guest loyalty and profile changes over time without losing historical data.

- **Logic:** When a record in customer_data.csv is updated (e.g., a change in membership level or customer_rating), the SSIS SCD transformation expires the old record and inserts a new active row.
- **Audit Columns:** * RowStartDate & RowEndDate: Define the validity period of the record.
    - IsCurrent: A boolean flag (1 for active, 0 for historical) used to filter the most recent guest profile in Power BI.

5.3 Data Flow and Transformation Logic

The FactBookings Data Flow task is the most complex component, as it merges 37 columns from three heterogeneous sources.

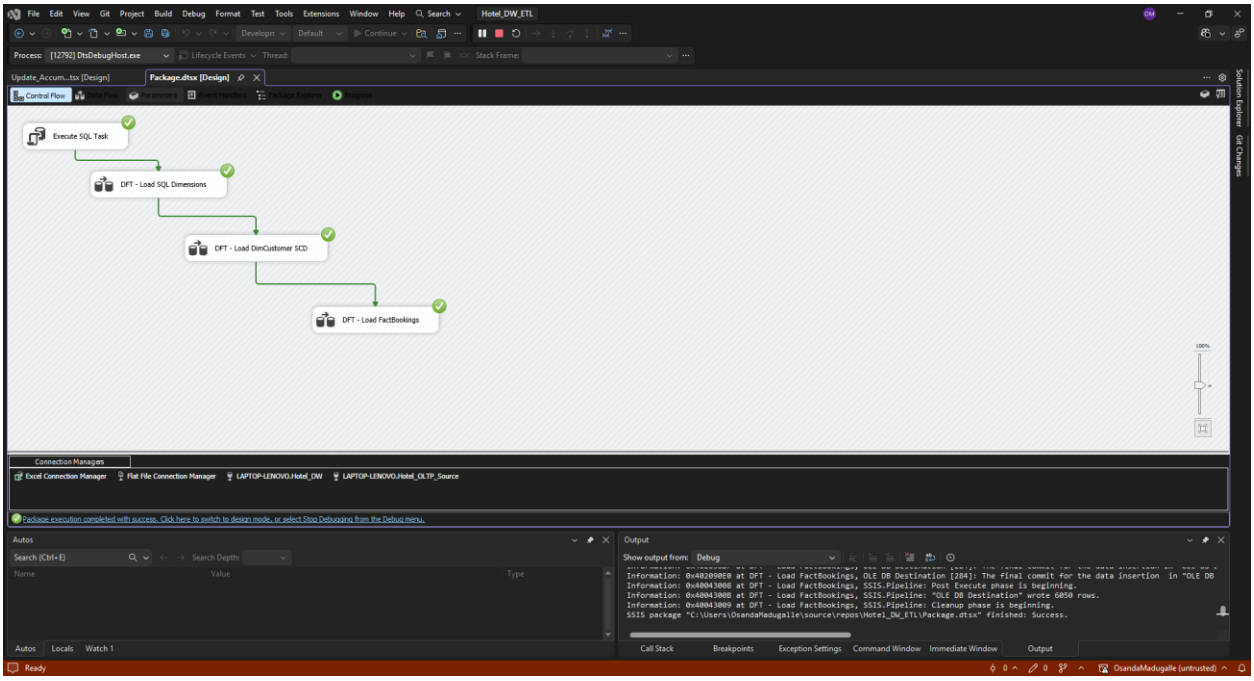
Key Transformations Used:

- **Lookups:** Crucial for transforming Natural Keys from Excel and CSV into Surrogate Keys (SK) from the Warehouse. For example, hotel_id is looked up against DimHotel to retrieve the HotelKey.
- **Data Conversion:** Since the source files (Excel/CSV) often import data as Unicode String, we used Data Conversion transformations to cast metrics like gross_amount and net_sales into Numeric and Currency formats compatible with SQL Server.
- **Derived Columns:** Used to create new business logic at the ETL level:
    - net_sales: Calculated as [gross_amount] - [discount_amt].
    - txn_process_time_hours: Calculated using the DATEDIFF function between accm_txn_create_time and accm_txn_complete_time.

5.4 Handling Heterogeneous Sources

To meet the project requirements, the ETL package manages three different connection managers:

1. **OLE DB Connection:** Connects to the Hotel_OLTP_Source for relational tables.
2. **Excel Connection Manager:** Extracts 6,050 rows from booking_data.xlsx.
3. **Flat File Connection Manager:** Parses the customer_data.csv for guest details.



	BookingKey	HotelKey	RoomKey	SalespersonKey	CustomerKey	ArrivalDateKey	DepartureDateKey	resv_status	nights	adult	child	dis_channel	payn
1	1	3	2	1	1	20240402	20240410	Check-Out	8	2	0	Direct	Crec
2	2	3	1	2	2	20240713	20240719	Check-Out	6	2	0	Direct	Crec
3	3	3	2	3	3	20240822	20240904	Check-Out	13	1	0	Agoda	Onli
4	4	1	4	1	4	20240516	20240522	Check-Out	6	3	1	Direct	Crec
5	5	2	1	1	5	20240205	20240210	Check-Out	5	2	2	Direct	Onli
6	6	3	2	2	6	20240913	20240927	Check-Out	14	1	0	Direct	Crec
7	7	2	2	4	7	20240912	20240917	Check-Out	5	1	0	Agoda	Onli
8	8	1	2	5	8	20240915	20240930	Check-Out	15	1	2	Booking.com	Onli
9	9	2	1	1	9	20240624	20240705	Check-Out	11	2	0	Booking.com	Onli

Task 06: ETL Development – Accumulating Fact Tables

6.1 Concept of the Accumulating Snapshot Fact Table

In dimensional modeling, an **Accumulating Snapshot Fact Table** is used to track the progress of a business process that has a defined start and end point. Unlike a periodic snapshot, which records state at a single moment, this model is updated multiple times as a transaction moves through various milestones.

For this project, the FactBookings table was designed to monitor the lifecycle of a hotel reservation, capturing the duration between the initial system entry and the final transaction completion.

6.2 Fact Table Schema Extension

To support this requirement, the FactBookings table was extended with three specific columns. This allows for the storage of both timestamps and the resulting performance metric.

Column Name	**Data Type**	**Purpose**
accm_txn_create_time	DATETIME	Captured during the initial ETL load via the source system.
accm_txn_complete_time	DATETIME	Populated during the secondary "Update" ETL process.
txn_process_time_hours	INT	A calculated measure of the total processing duration.

SQL Implementation:

SQL:

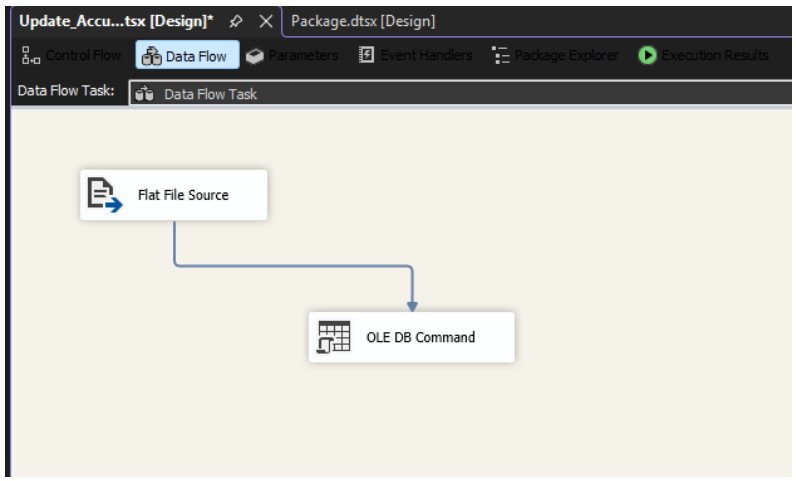
```
ALTER TABLE FactBookings
ADD accm_txn_create_time DATETIME,
    accm_txn_complete_time DATETIME,
    txn_process_time_hours INT;
```

6.3 Implementation of the Update Pipeline

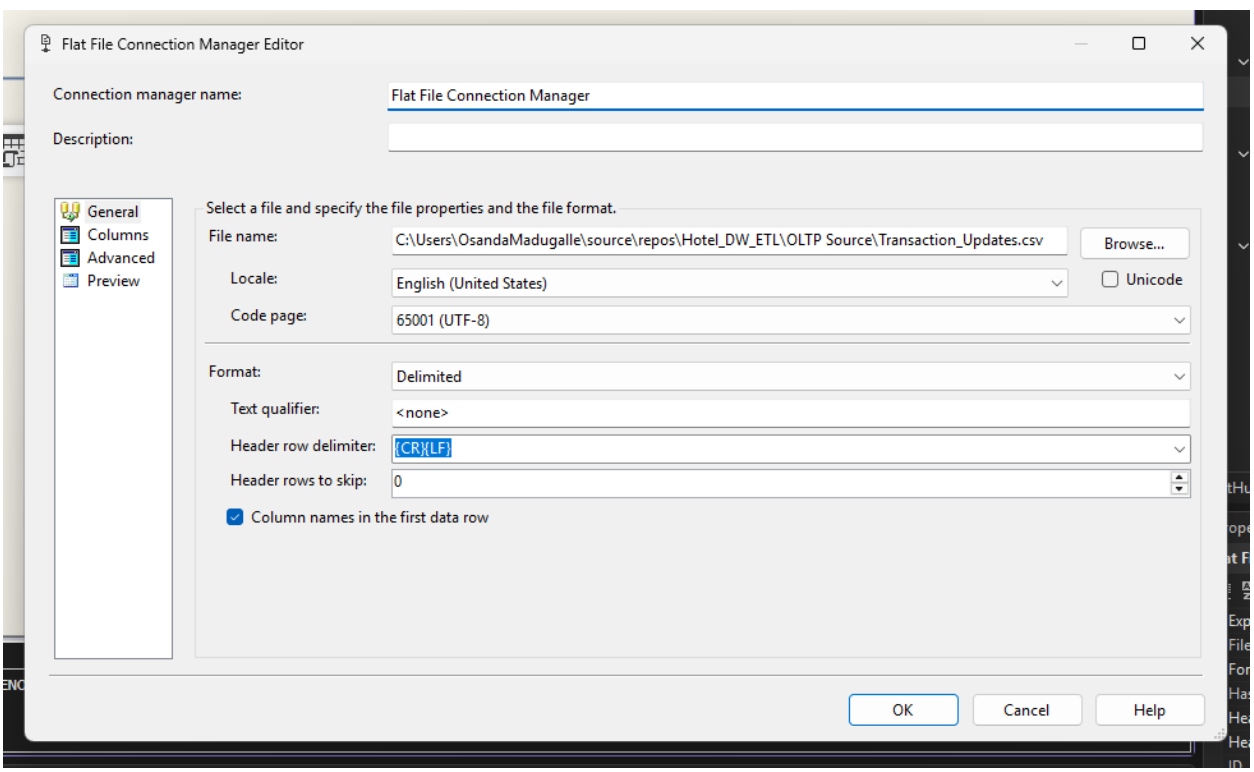
The update logic was developed in a dedicated SSIS package titled **Update_Accumulating_Fact.tsx**. This package follows a two-stage execution pattern to ensure data precision.

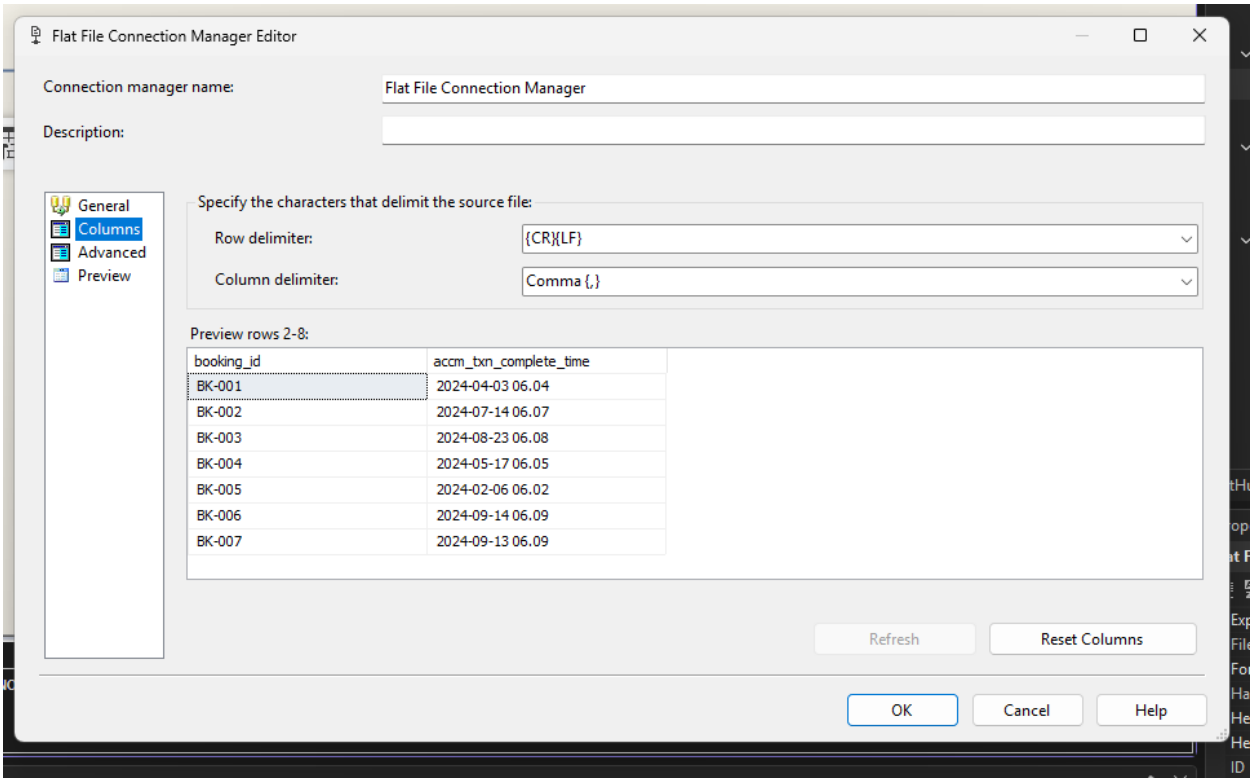
Stage 1: Row-Level Updates (Data Flow)

The Data Flow Task extracts finalized transaction data from an external source (Transaction_Updates.csv) and maps it to the existing fact records.

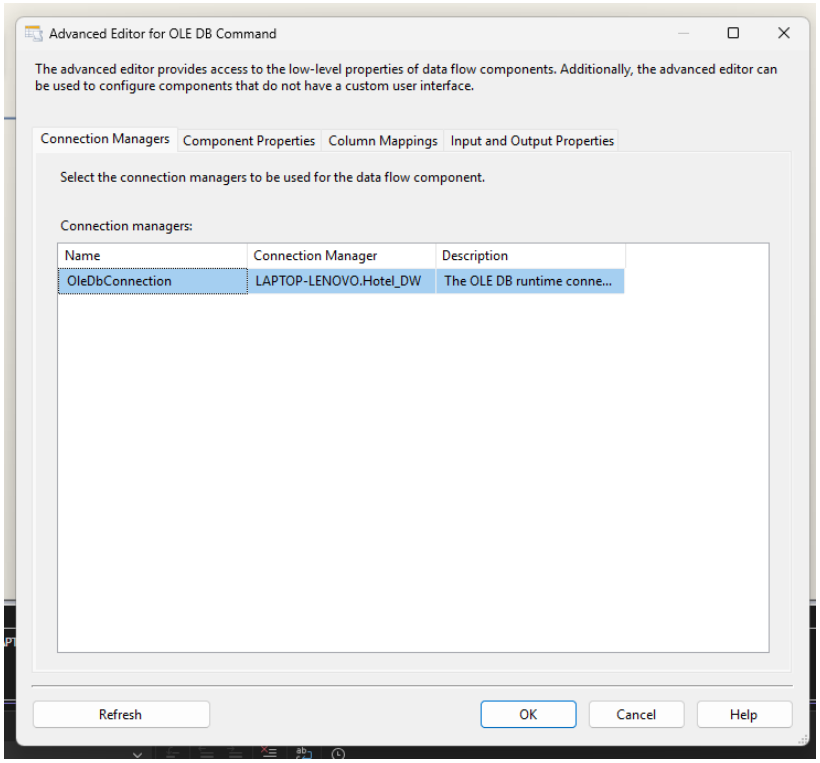


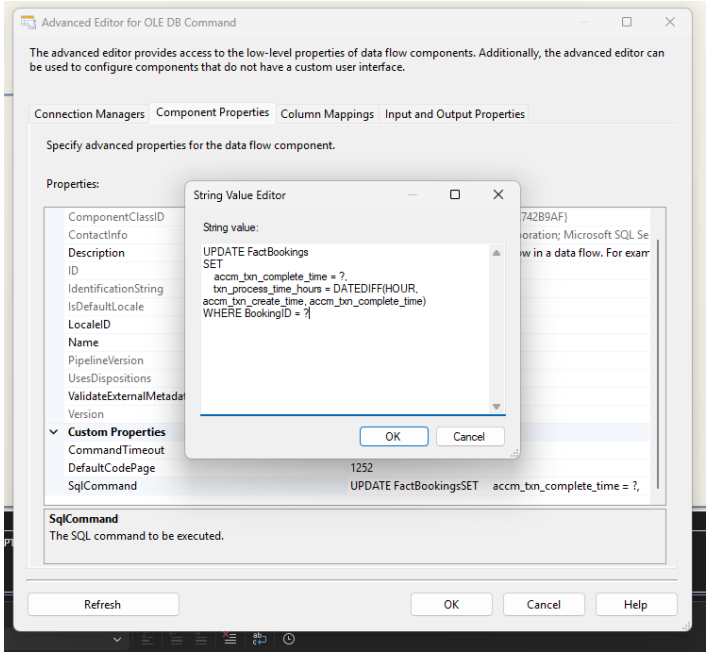
1. **Flat File Source:** Reads the BookingID and the accm_txn_complete_time.





2. **OLE DB Command:** Executes a row-by-row update. This command maps the completion time and performs the initial duration calculation.

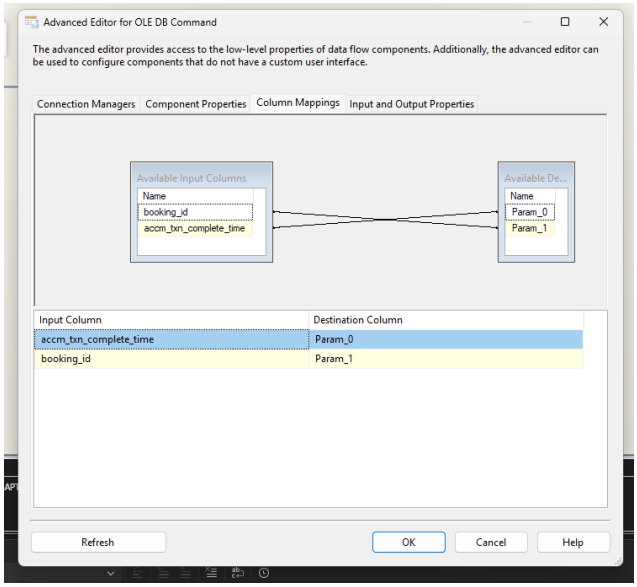




SQL Command used in OLE DB Command:

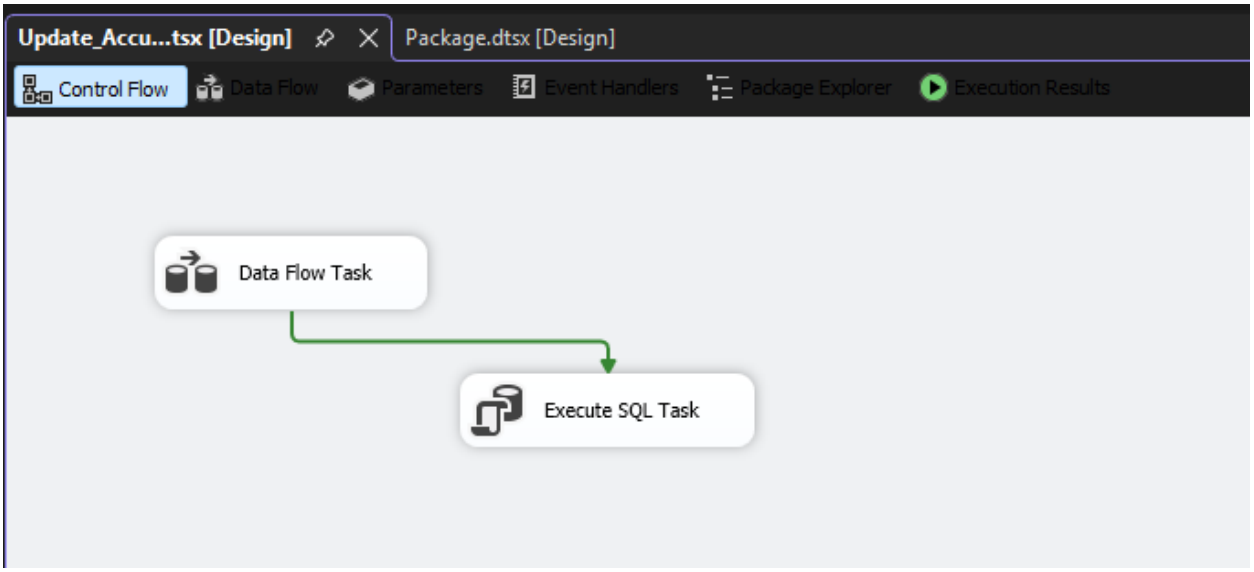
SQL:

```
UPDATE FactBookings
SET
   accm_txn_complete_time = ?,
   txn_process_time_hours = DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time)
WHERE BookingID = ?;
```



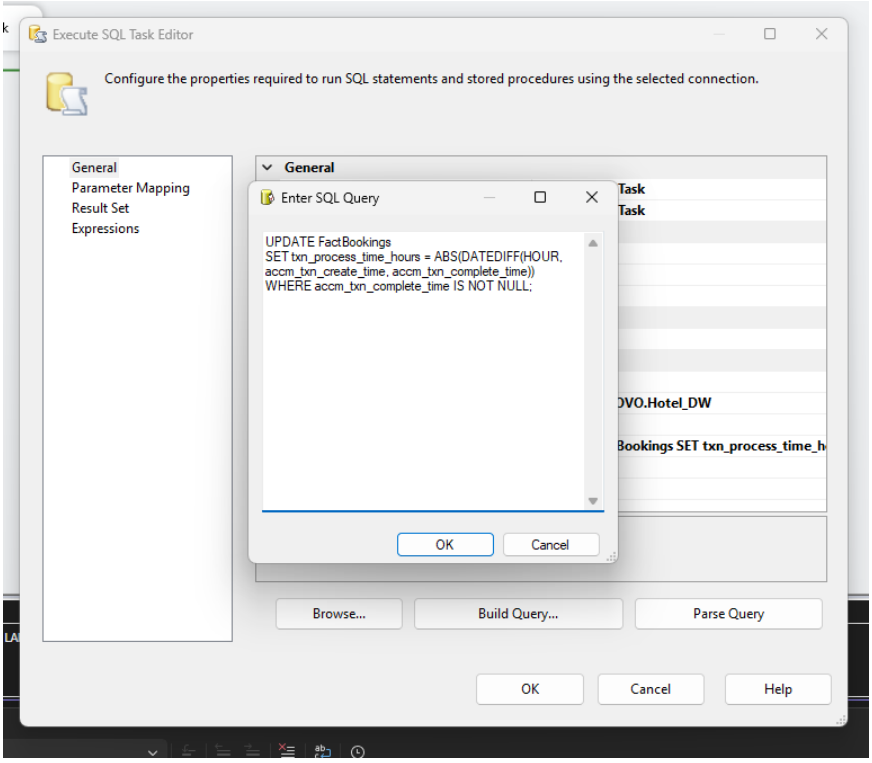
Stage 2: Batch Integrity & Validation (Control Flow)

Following the Data Flow, an **Execute SQL Task** was implemented in the Control Flow to act as a data integrity layer.



This task performs a set-based update on all completed records. The use of the ABS() function is a critical design choice to ensure that txn_process_time_hours is always a positive integer, even in cases of minor system clock discrepancies between different sources.

Final Cleanup Script:

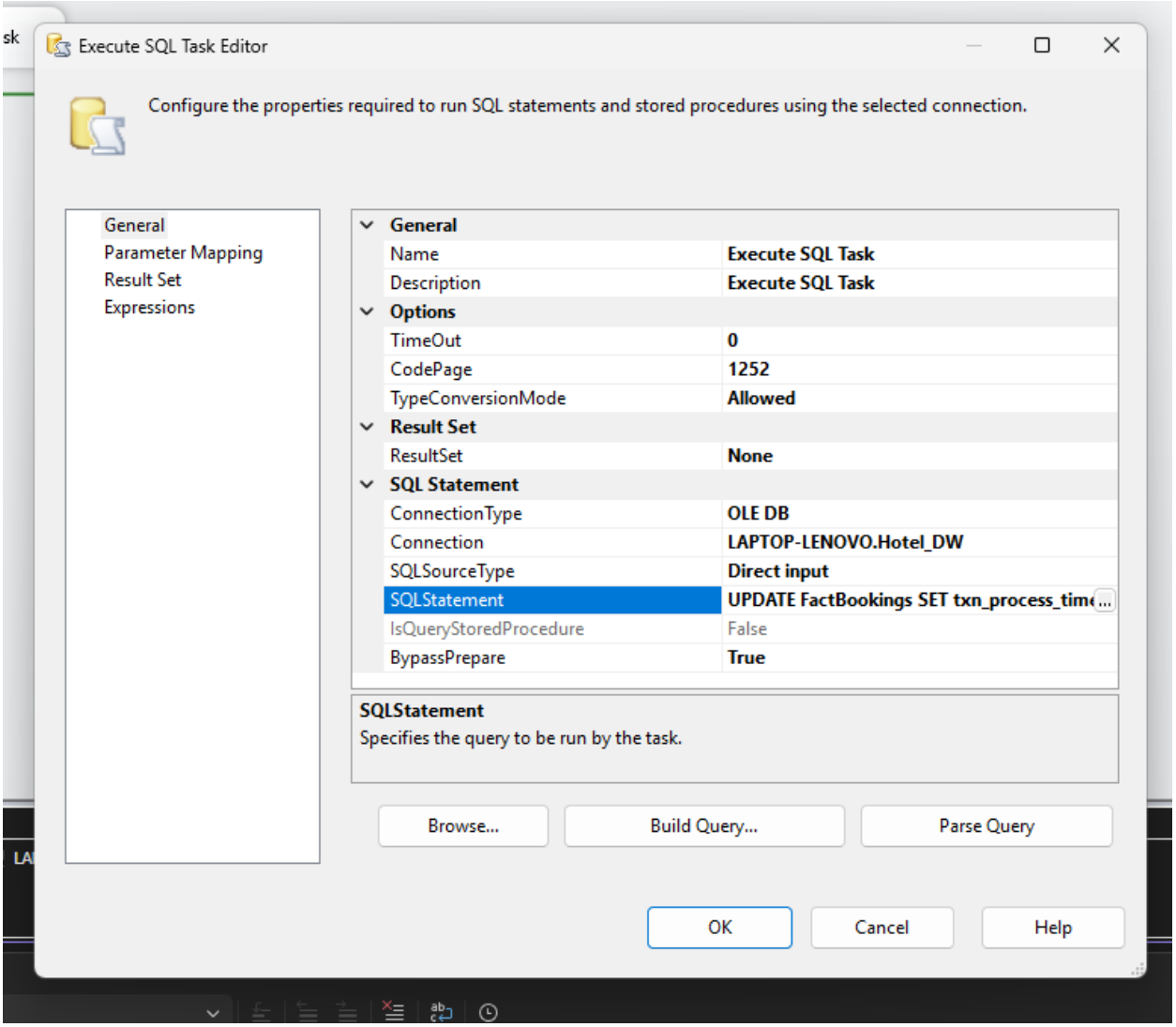


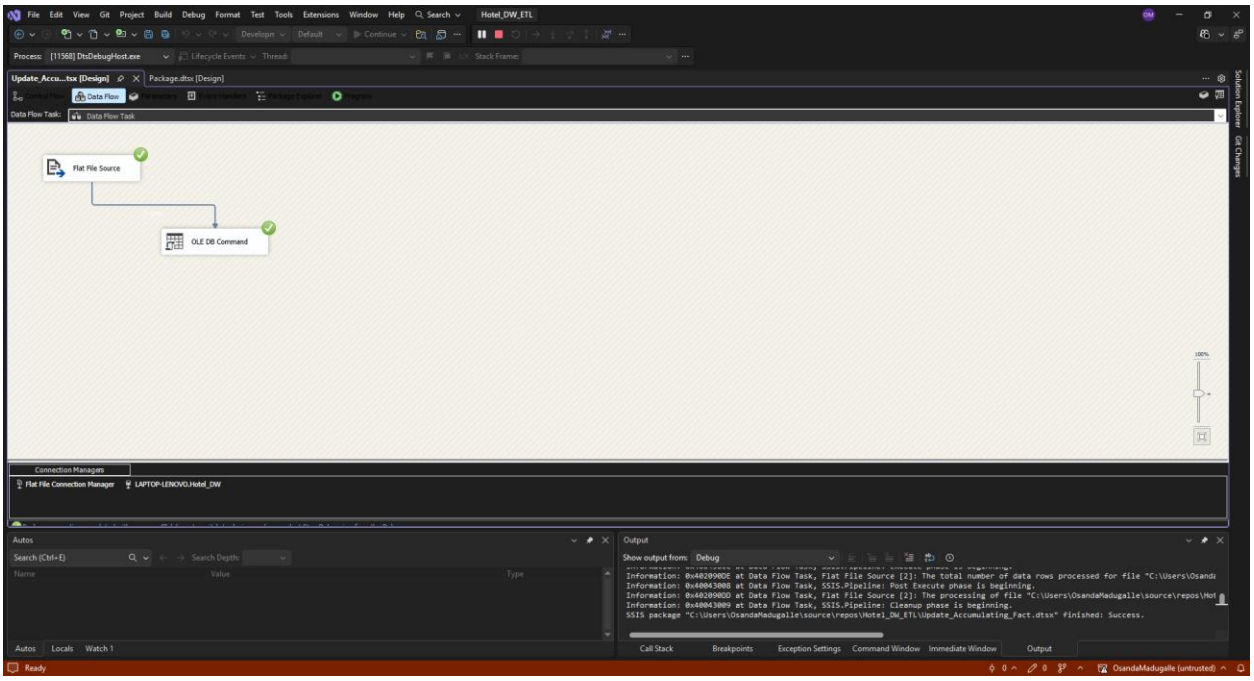
SQL:

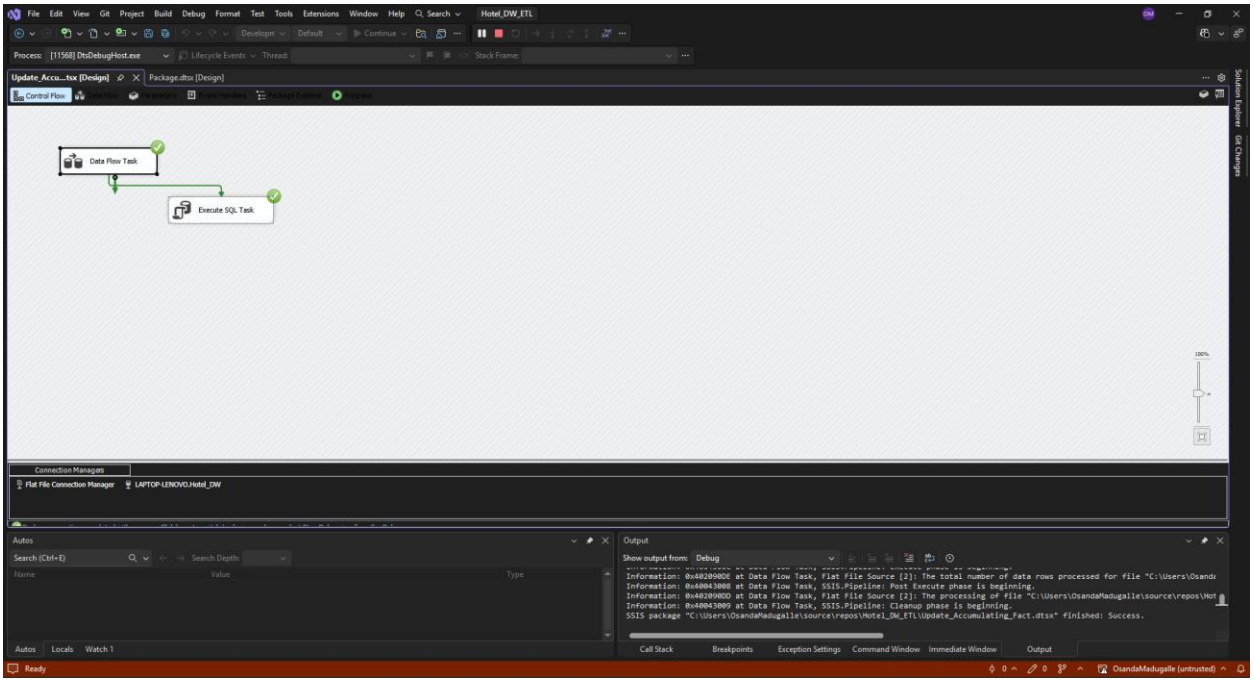
UPDATE FactBookings

SET txn_process_time_hours = ABS(DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time))

WHERE accm_txn_complete_time IS NOT NULL;







6.4 Data Validation and Testing

	e_per_night	arrival_date_time	departure_date_time	stay_duration_days	meal	acom_txn_create_time	acom_txn_complete_time	txn_process_time_hours	BookingID
1	.00	2024-04-02 00:00:00.000	2024-04-10 00:00:00.000	8	BB	2024-04-02 00:00:00.000	2024-04-03 06:04:00.000	30	BK-001
2	.00	2024-07-13 00:00:00.000	2024-07-19 00:00:00.000	6	BB	2024-07-13 00:00:00.000	2024-07-14 08:07:00.000	32	BK-002
3	.00	2024-08-22 00:00:00.000	2024-09-04 00:00:00.000	13	BB	2024-08-22 00:00:00.000	2024-08-23 10:08:00.000	34	BK-003
4	0.00	2024-05-16 00:00:00.000	2024-05-22 00:00:00.000	6	BB	2024-05-16 00:00:00.000	2024-05-17 11:05:00.000	35	BK-004
5	.00	2024-02-05 00:00:00.000	2024-02-10 00:00:00.000	5	BB	2024-02-05 00:00:00.000	2024-02-06 13:02:00.000	37	BK-005
6	.00	2024-09-13 00:00:00.000	2024-09-27 00:00:00.000	14	BB	2024-09-13 00:00:00.000	2024-09-14 15:09:00.000	39	BK-006
7	.00	2024-09-12 00:00:00.000	2024-09-17 00:00:00.000	5	BB	2024-09-12 00:00:00.000	2024-09-13 06:09:00.000	30	BK-007